

June 19, 2014

Inflow and Infiltration Allowance Assessment

Integrated Liquid Waste and Resource Management

A Liquid Waste Management Plan for the Greater Vancouver
Sewerage & Drainage District and Member Municipalities



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Abbreviations

AAF	Average annual flow
ADWF	Average dry weather flow
BMP	Best management practice
CCME	Canadian Council of Ministers of Environment
CCTV	Closed circuit television
CSO	Combined sewer overflow
CWS-MMWE	Canada-wide Strategy for the Management of Municipal Wastewater Effluent
DONCE	Discharge out of the normal course events as pertaining to the Fisheries Act
ENSO	El Niño Southern Oscillation
GVS&DD	Greater Vancouver Sewerage and Drainage District
GDI&I	Groundwater derived inflow and infiltration
I&I	Rainwater inflow and groundwater infiltration, expressed in this report as flow per area (L/ha-d)
I&IMP	Inflow and infiltration management plan
IDF	Intensity, duration, frequency: statistical graphs for rainfall
ILWRM	Integrated Liquid Waste and Resource Management: A Liquid Waste Management Plan for the Greater Vancouver Sewerage & Drainage District and Member Municipalities
IPREM	Integrated Partnership for Regional Emergency Management in Metro Vancouver
LWMP	Liquid Waste Management Plan
PDO	Pacific Decadal Oscillation
PDWF	Peak dry weather flow
PWWF	Peak wet weather flow
PEP	Provincial Emergency Program

RDI&I	Rain derived inflow and infiltration
REAC	Regional Engineering Advisory Committee
REAC-LWSC	Liquid Waste Subcommittee of the Regional Engineering Advisory Committee
SCADA	Supervisory control and data acquisition
SSO	Sanitary sewer overflow
WSER	Wastewater Systems Effluent Regulations (under the Fisheries Act)
WWTP	Wastewater treatment plant

Summary

Working with the Regional Engineering Advisory Committee's Liquid Waste Sub-committee, Metro Vancouver staff assessed the current regional inflow and infiltration allowance of 11,200 L/ha-d. Metro Vancouver considers inflow and infiltration rates above this amount to be excessive. This allowance is used to determine the basic service capacity of Metro Vancouver's trunk sewers and in turn inform if new connections can be accommodated as well as when capacity upgrades will be required. As unmanaged, excessive inflow and infiltration consumes the sewer capacity invested for future community growth, it can consequently trigger avoidable sewer capacity upgrades and can result in reoccurring sanitary sewer overflows.

As part of the assessment, previous inflow and infiltration management business cases were examined and a new evaluation completed. From both analyses, inflow and infiltration management was determined to be more cost effective than conveying and treating inflow and infiltration in excess of the 11,200 L/ha-d design allowance. In addition, inflow and infiltration allowances and mitigation experience from the Metro Vancouver region and other jurisdictions in North America, Europe and Australia were also examined.

Review of inflow and infiltration reduction programs confirms that controlling inflow from private laterals is generally required to achieve significant reductions in inflow and infiltration rates. Sewer use bylaw enforcement and private lateral quality inspection play important roles in the long-term reductions in inflow and infiltration, particularly as neighbourhoods are redeveloped.

A key finding of the assessment is the identification of semi-combined sewers—sanitary sewers with permitted foundation drain connections. Such systems were the historic norm in most of Canada and are usually referred to as sanitary sewers even though they respond very rapidly to rainfall, have high rates of inflow and infiltration and exhibit similar hydrologic characteristics as fully combined sewers. Semi-combined sewers have been identified as being built in the region; however, the full extents of these systems and their impacts have not been assessed.

Furthermore, the inclusion of private-side sewer laterals in municipal I&I programs is a critical component to successful long-term I&I management. One key challenge identified is determining the level when investments in I&I reduction no longer provide satisfactory returns. Given the range of existing I&I rates and diversity of factors between sewer catchments, it was not possible to determine any investment thresholds.

In consideration of the assessment of the I&I allowance in this report, the following recommendations are made:

1. That GVS&DD members use the "Envelope Method" as described in the 2011 *I&I Management Plan Template* to define I&I

2. That GVS&DD members share through Metro Vancouver the information they have learned implementing their I&I management plans to help ascertain if there are I&I reduction thresholds for old sewer catchments
3. That GVS&DD members work with Metro Vancouver to determine whether semi-combined sewers (sanitary sewers designed to accept foundation drain connections) are a significant factor affecting I&I rates and identify options to remedy semi-combined sewers
4. That, for the interim, Metro Vancouver continues to use 11,200 L/ha-d as the I&I allowance for basic service modeling for GVS&DD sanitary sewers and for capacity upgrades
5. That GVS&DD members will prioritize I&I management in sanitary sewer catchments tributary to reoccurring SSOs
6. That GVS&DD members will prioritize sanitary sewer catchments with the highest I&I rates for I&I reduction and mitigation
7. That this report and recommendations should be reviewed in four years

Purpose

The updated Liquid Waste Management Plan (Metro Vancouver, 2010), requires the review and evaluation of the rainwater inflow and groundwater infiltration (I&I) allowance into sanitary sewers as follows:

Action 1.1.9 Metro Vancouver will work with municipalities to review historical data and adjust as necessary the average inflow and infiltration allowance for regional trunk sewers and wastewater treatment plants, and develop associated target allowances for municipal sewer catchments associated with a 1:5 year return frequency storm event for sanitary sewers to a level that ensures environmental and economic sustainability.

The I&I allowance is an important factor in determining when Metro Vancouver's sewerage infrastructure requires capacity upgrades; however, achieving the allowance in Metro Vancouver's sewers is dependent upon the effectiveness of municipal I&I management programs.

At the February 16, 2012 meeting of the REAC-LWSC, the Sub-committee endorsed Metro Vancouver's approach to examine the impacts on the regional system if the I&I allowance were to double from 11,200 L/ha·d to 22,400 L/ha·d. The rationale for assessing a doubling of the current I&I allowance was based on observations by some municipalities that their I&I programs have not reduced I&I below the 25,000 L/ha·d range. As well, I&I rates for other jurisdictions¹, such as 22,400 L/ha·d used by Toronto (Chessie, 2011), supported the desire to review doubling the allowance.

This assessment was expanded to provide additional context and considered the I&I estimates provided as part of the 2013 ILRWM Biennial Report².

¹ However, higher I&I allowances such as 22,400 L/ha·d accounts for the past practice of connecting building foundation drains to sanitary sewers which was common practice in Ontario (Parhami, 2013) and other provinces (Darbyshire, 2013) until around the late 1970s to early 1980s.

² Municipal and regional I&I estimates (Liquid Waste Services Department, 2013a; Liquid Waste Services Department, 2013b)

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Background

As part of the development of the 2002 Liquid Waste Management Plan, Metro Vancouver evaluated the current I&I allowance of 11,200 L/ha·d and its use in the calculation of basic service flow for sewer capacity analysis. Under Action 1.2.4, the Liquid Waste Management Plan also requires the prevention of wet weather sanitary sewer overflows (SSOs) associated with 24 hour storm events of magnitudes less than those with a 1:5 year return frequency (Metro Vancouver, 2010, p. 17). Thus, the I&I allowance for sewer capacity analyses is linked to 1:5 year return storms.

While some associate the I&I allowance with the Rawn Report, there is no explicit I&I allowance included in this report (Rawn, Hyde, & Oliver, 1953, p. 132). Although the allowance converts to 1000 Imperial gallons per acre per day which is suggestive of a design rule, the allowance was adapted by the GVS&DD from sewer design standards developed by Seattle³ and is comparable with the current 11,300 L/ha·d I&I design allowance used in King County (EarthTech Inc., 2003; City of Bellevue, 2013). The allowance was subsequently evaluated as part of the region-wide I&I assessment work in the 1990s.

Region-wide I&I Assessment 1991-1999

In 1991 Metro Vancouver and GVS&DD members started a comprehensive assessment of I&I within the region. This included basic assessments of municipal I&I and municipal sewer construction requirements.

By 1995, municipal-regional technical work groups for Inflow and Infiltration Analysis, Sewer Systems Evaluation Surveys and New Construction had completed their investigation and produced reports under the Inflow and Infiltration Reduction Program.

At the time, some GVS&DD members reported I&I rates greater than the 11,200 L/ha·d while others reported meeting the allowance. Although some areas had I&I rates much greater than the target, for example Figure 1 (North Shore Sewerage Area Technical Advisory Committee, 2000, p. 7), the target appeared attainable given that there were also catchments with much lower I&I rates as shown by Figure 2 and Figure 3 (Fraser Sewerage Area Technical Advisory Committee, 2000, p. 31).

Subsequently, financial assessments were carried out on the Fraser, Lulu Island and North Shore Sewerage Areas comparing the capital and net present value of different options for asset management, I&I management and system conveyance capacity⁴. This was accompanied by detailed I&I assessment programs for the Fraser and North Shore Sewerage Areas. The results were used to form the strategies of the 2002 LWMP where targeted I&I management programs were determined to be more cost effective than conveying and treating excessive I&I.

³ The I&I allowance of 11,200 L/ha·d was adopted during the Rawn Report Update in the unpublished 1972 report.

⁴ See Appendix A: 1999 I&I Business Cases.

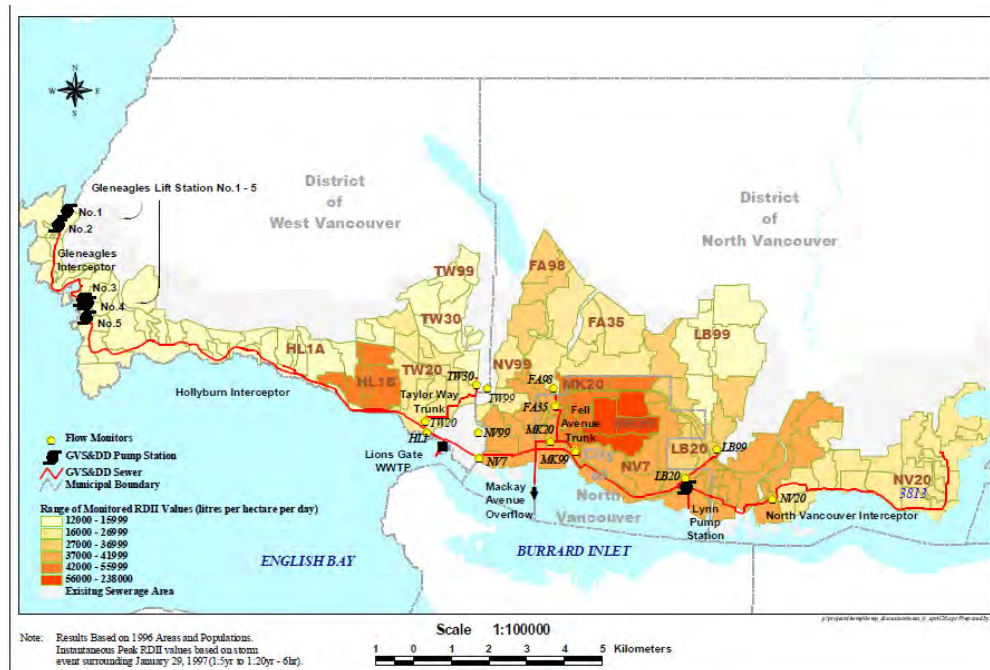


Figure 1 Inflow and Infiltration North Shore Sewerage Area in 1999

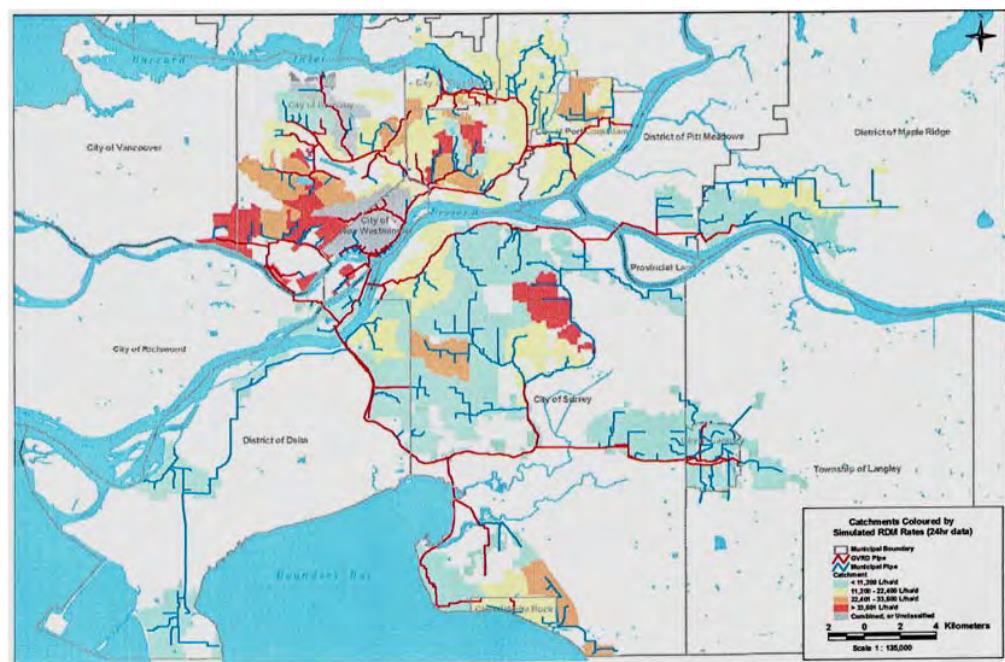


Figure 2 Inflow and Infiltration Fraser Sewerage Area in 1999

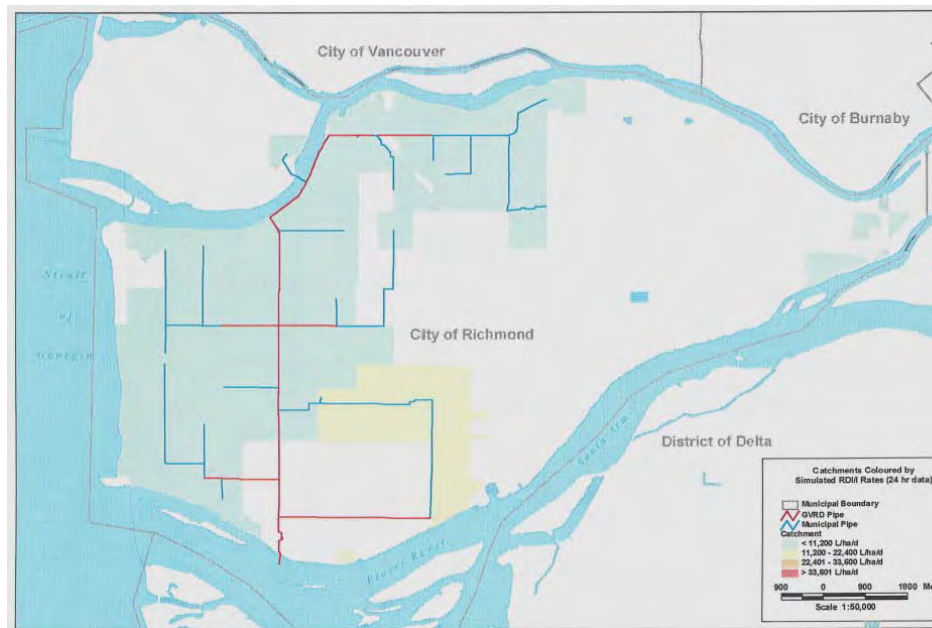


Figure 3 Inflow and Infiltration Lulu Island Sewerage Area in 1999

2002 Liquid Waste Management Plan

The 2002 Liquid Waste Management Plan was supported by technical work which includes business cases for I&I management which were developed by a joint municipal-regional task groups to address the North Shore Sewerage Area (North Shore Sewerage Area Technical Advisory Committee, 2000), the Lulu Island West Sewerage Area, and the Fraser Sewerage Area (Fraser Sewerage Area Technical Advisory Committee, 2000).

These business cases are summarized in Appendix A: 1999 I&I Business Cases. This work determined that reducing excessive I&I conveyed in Metro Vancouver sewers to the allowance target of 11,200 L/ha-d would be less costly than constructing additional system capacity to manage excess wet weather flows. This formed the basis of the 2002 LWMP.⁵

In addition, in 2001, GVS&DD members agreed that they would manage their assets and private service laterals so as not to exceed the regional allowance of 11,200 L/ha-d. This strategy was endorsed by all GVS&DD member municipal councils and approved by the Minister of Environment in 2002. The policy and commitment statements for Infrastructure Management (Policy and Planning Department, 2002) are as follows:

⁵ See Appendix A: 1999 I&I Business Cases.

P8 Infrastructure Management

The District and its member municipalities will establish sewer system infrastructure management programs that will maintain the regional trunk and interceptors, the municipal collection system, and the private service laterals in a state of good repair. The objective will be to ensure the sustainability of the collection system so that expensive repair and rehabilitation is not deferred to future generations and that the average daily infiltration and inflow will not exceed 11,200 litres per hectare per day as a result of a storm with less than a five year return period.

C19 Infrastructure Management

The District and its member municipalities will establish ongoing sanitary sewer system evaluation programs to determine the condition of the regional trunk sewerage system, the municipal sewerage system, and private property service laterals. As required, legislative and legal authority will be sought to address infiltration and inflow originating from private property service laterals. These evaluation programs will be ongoing and determine the condition of the entire sewer system over a 20 year time cycle. The District and its member municipalities will develop and apply a consistent approach to sewer system evaluation surveys.

Repair and replacement programs will be established based on targets set for sanitary sewer overflow reduction and the severity of infiltration and inflow relative to the design allowance of 11,200 litres per hectare per day.

Implementing the I&I Management Options from the 2002 LWMP

In implementing the 2002 LWMP, Metro Vancouver and its members formed a task group to examine sewer condition reporting, with the scope of this work confined to inspection and assessment of regional and municipal sewers. Recommendations for a common, region-wide approach were summarized in a working draft report (EarthTech & Kerr Wood Leidal Assoc., 2002) which was subsequently disseminated to member municipalities.

Although LWMP Biennial Reports published by Metro Vancouver in 2004, 2006, 2008 and 2010 reported I&I estimated at Metro Vancouver flow meter sites, interpretation of the estimates was complicated as the estimates were for the cumulative upstream catchment. In addition, municipal I&I management efforts were not documented in the biennial reporting as only quantities of sewer asset inspection and rehabilitation were reported. As a result, awareness and sharing of municipal I&I management program progress and catchment I&I rates was limited.

Measuring I&I

The conventional methodology, as shown by Figure 4 (National Research Council & Federation of Canadian Municipalities, 2003), for estimating I&I provides I&I rates that are tied to the magnitude of

storm events recorded during the investigation and do not provide an associated projected I&I rate for the 1:5 y return storm event.

This approach does not allow for easy comparison of I&I values for different storm return periods or for different catchments. Attempting to provide a means to compare I&I rates among catchments and between storm events, an approach referred to as the Envelope Method was developed and introduced to Metro Vancouver through the development of *Sewer Condition Reporting and Assessment Template* (EarthTech & Kerr Wood Leidal Assoc., 2002).

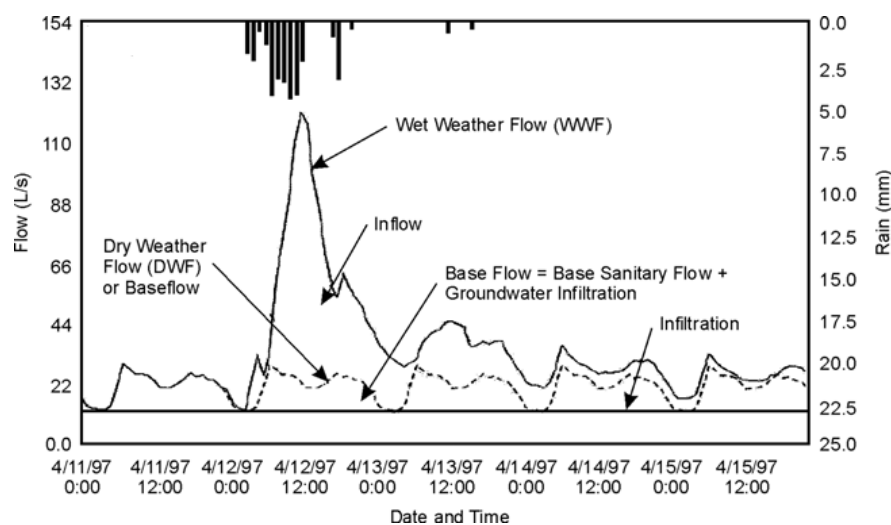


Figure 4 Generalized Illustration of I&I Components

The Envelope Method

The Envelope Method is included in the *Inflow and Infiltration Management Plan Template*, developed for the REAC-LWSC, as the recommended methodology for estimating municipal I&I (Kerr Wood Leidal Assoc., 2011). This method estimates I&I through the extrapolation of correlated rain and sewer flow events.

In using the I&I Envelope Method, it is noted that:

...“smaller catchments, or catchments with larger inflow components will tend to correlate [RDI&I] better with shorter rainfall durations; [and] larger catchments, or catchments with less inflow and more infiltration will correlate [RDI&I] better with longer rainfall durations” (Kerr Wood Leidal Assoc., 2011).

This observation indicates that catchment size and hydrologic time of concentration play a role in determining the maximum I&I flows. As with hydrology, small catchments with short times of concentration tend to have greater peak to average flow ratios. If I&I behaves similarly, then I&I rates may be assumed to be estimated higher for small catchments as compared to larger ones, if sewer condition and other factors are equal.

The Envelope Method relies upon fitting and extrapolating a curve for precipitation and I&I. The curve is then shifted upwards to represent fully saturated groundwater conditions. The correlation, the accuracy of the data, validity of shifting the corresponding antecedent moisture conditions to fully saturated ground, and extrapolation of storm return periods all introduce the potential for error.

Underlying this approach are I&I estimates which can also be affected by the accuracy of data collection equipment, resolution of time step, and the magnitude of the return periods of rain captured during the monitoring period.

As a result, it is felt that the comparison of I&I estimates developed using the Envelope Method should be treated somewhat broadly, particularly when making comparisons between catchments.

I&I Estimates and Pump Stations

Sewage pump stations offer a means to provide flow measurement. A catchment hydrograph can be developed with its quality depending upon the type of pump station and type of data logging of pump run times and flows. However, the pump capacities as well as wet well sizing relative to flow rates will determine how much peak flow attenuation is created by the pump station. As a result, I&I estimated using pump station run-time data is not the actual influent I&I, but the effective I&I. The difference between actual influent I&I and effective I&I is pump station specific.

Precipitation Considerations

Volume of Rainfall and Return Periods

Through the ILWRM's definition of basic service, the regional sewer I&I allowance is linked to the reoccurrence threshold for wet weather SSOs: a 1:5 year return storm event with a 24 hour duration⁶. Furthermore, setting a 24 hour storm duration does not reflect the actual hydrologic times-of-concentration nor the critical storm event for all catchments, particularly smaller municipal sewer catchments.

Annual precipitation varies widely across the region. As the I&I allowance is linked to the 24 hour, 1:5 year return storm event, the total volume of rainfall associated with this event also varies across the region, as shown by Figure 5.

While the lowest 24 hour, 1:5 year return rainfall volumes, between 55 to 65 mm, are clustered around Steveston, Ladner and Tsawwassen, other areas sharing similar rainfall volumes are broadly dispersed such as 65 to 70 mm areas at UBC, South Vancouver and White Rock. The highest rainfall volumes, greater than 100 mm, are along the mountains from North Vancouver District to Maple Ridge, including Burnaby Mountain.

⁶ per ILWRM Action 1.3.2 which defines basic service and Action 1.1.9 regarding SSOs strategies (Metro Vancouver, 2010, pp. 17, 18)

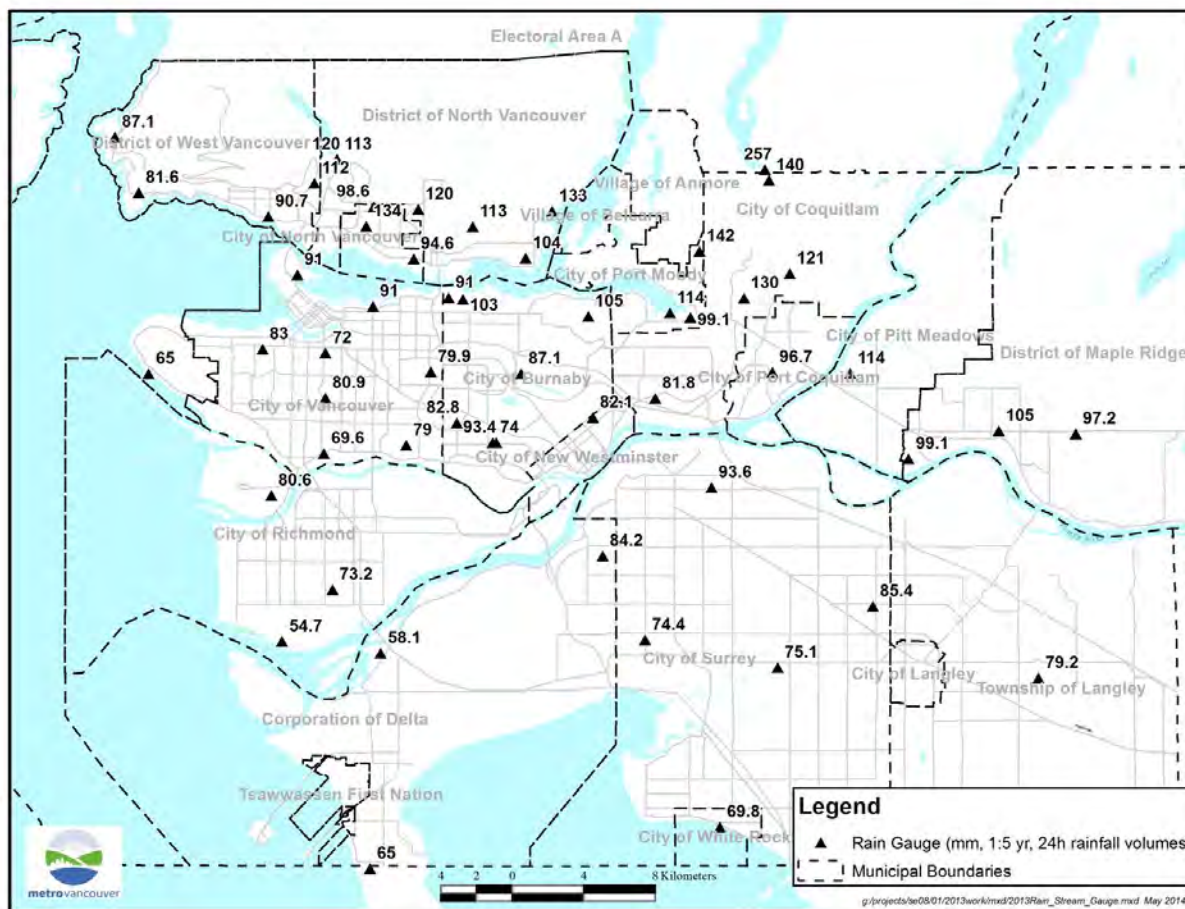


Figure 5 Total Rainfall (mm) for 24 h Duration, 1:5 y Return Storm Event

Considering the variations in I&I among catchments sharing similar total rainfall is significant, I&I rates do not appear to be linked to total rainfall. For example, there are catchments in Maple Ridge with 100 to 105 mm of rainfall and in West Vancouver with 80 to 90 mm of rainfall which either meet or come close to the 11,200 L/ha-d I&I allowance. In contrast, there are other catchments in Burnaby, White Rock, and Coquitlam where rainfall is similar or less than 100 mm, but these other catchment have much higher I&I rates⁷.

Climate Change and Cycles

British Columbia's climate is heavily influenced by climate oscillation: the dominant PDO and the weaker ENSO. The PDO is divided into warm and cool phases typically spanning several decades, with each phase having its own higher or lower IDF pattern. While IDF curves continue to be used as part of

⁷ See Figure 7 and Figure 8 of this report for regional catchment I&I estimates and Figure 10 for municipal I&I estimates.

standard hydrology practice, they do not reflect the distinctiveness and nuances of the different PDO rainfall intensities of the different phases.

The PDO shifted from a warm to an unstable cool phase around 1998. The ENSO being weaker with a more frequent oscillation, also influences precipitation intensity and compounds with the effects of the PDO. As of 2014, the PDO is back in a cold phase. However, observations between 2000 and 2014 indicate instability of the PDO cycle by the presence of much shorter phases. It has been suggested that the normal PDO cycle has broken down, possibly as a result of climate change (National Oceanic and Atmospheric Administration, 2014).

Considering the influence of climate oscillations, I&I monitoring undertaken in a cool, dry phase and will not be likely be reflective of conditions in the wet phase when rainfall intensity is expected to be greater.

Depending on the PDO and ENSO phases when the IDF data was collected, upper and lower IDF curves can be generated for each rain gauge station if the data are split into their wet and dry phases. Such a split was observed at Metro Vancouver's Coquitlam QT10 rain gauge station in 2000 for short duration storms. Two IDF curves were produced for each PDO phase with the 1:10 year dry IDF curve being very similar to the 1:5 wet PDO phase IDF curve.

These observations initiated further detailed investigation on precipitation as related to climate change and climate oscillation impacts. In consideration of known climate oscillations and analyzing climate change models and local rainfall data, future scenarios project that the IDF return period will approximately halve by 2050 (BGC Engineering Inc., 2009a). This may be interpreted as the current 1:10 year return storm event will become the future 1:5 year return storm event in 2050. Together, increased rainfall from climate change along with PDO and ENSO phase shifts will impact I&I rates. (Kerr Wood Leidal Assoc., 2011, p. 3 of Sec. 4). A series of projected IDF curves for future climate were prepared for Metro Vancouver (BGC Engineering Inc, 2009b), these may be useful for long-term I&I strategies.

By mid-century, the 24 hour, 1:5 year return storm rainfall intensity is projected to increase approximately 3% to 5% beyond the current rate (BGC Engineering Inc., 2009a); it is not clear if this increase necessitates any upward adjustment of the Metro Vancouver I&I allowance. For catchments with relatively short times-of-concentration, future rainfall intensities may be a concern as rainfall intensities for the 1 hour duration, 1:5 year storm are projected to increase around 17 to 18% by 2050 (BGC Engineering Inc., 2009a).

State of I&I in the Region

I&I in Regional Sewers

Metro Vancouver has not undertaken an I&I assessment of its sewers. The majority of the I&I is believed to originate from municipal sewers, similar to the experience in King County system where 95% of its I&I originates from local sewers⁸ (Wastewater Treatment Division, 2004, p. 1).

The importance of addressing I&I from private sewers is highlighted by King County where it is reported that approximately 50% of I&I is estimated to originate from private sewer laterals⁹ (Lampard, 2003). This is consistent with the broad range of 20 to 80% of total I&I originating from private sewers reported by local US governments (Strand Associates Inc., 2006, p. 2).

I&I in Municipal Sewers

During the review and updating of the LWMP in 2008 and 2009, some municipalities reported difficulties limiting the I&I rate to 11,200 L/ha-d. This indicated that there is a lack of knowledge as to why I&I rates vary so widely across the region, and as to why some municipalities have been successful in matching the I&I allowance while others have not, even after significant investment in sewer rehabilitation. Consequently, an action to review I&I allowance was included in the updated LWMP.

Figure 6 shows the results of I&I rehabilitation work as determined for the City of Burnaby for both mainline sewer and public-side lateral rehabilitation. This figure suggests that an I&I threshold of 22,400 L/ha-d is more realistic for this catchment than an allowance of 11,200 L/ha-d.

However, Burnaby's rehabilitation work was limited to public rights-of-way. Private sewer lateral rehabilitation was not undertaken (Boyland & Weismiller, 2011). Considering the lowest reported I&I rates were around 22,400 L/ha-d, this rate has been suggested as a possible I&I threshold when private lateral derived I&I is not directly mitigated.

In addressing I&I reduction progress by GVS&DD members over the last 10 years, I&I values reported in the Biennial Report: 2010-2012 (Liquid Waste Services Department, 2013b) show a continued wide variation between municipalities. For municipal catchments, reported I&I varies from below 11,200 L/ha-d to more than 100,000 L/ha-d. These variations may be due to:

- field changes over many decades such as tie-ins and disconnections may increase infiltration points;
- disparities between current municipal standards and the quality of historical sewer lateral construction;
- historical connection of foundation drainage to sanitary sewers (which was once common practice in the rest of Canada);
- differences among municipalities with their sewer asset management programs and budgets;

⁸ Local sewers include municipal and private sewers.

⁹ Referred to as private-side sewers.

- variation between municipalities with their sewer use bylaw enforcement with respect to cross-connections;
- differences between the age and materials of the sewers and the sewer laterals within and between communities;
- cross-connections due to “do-it-yourself” home repairs.

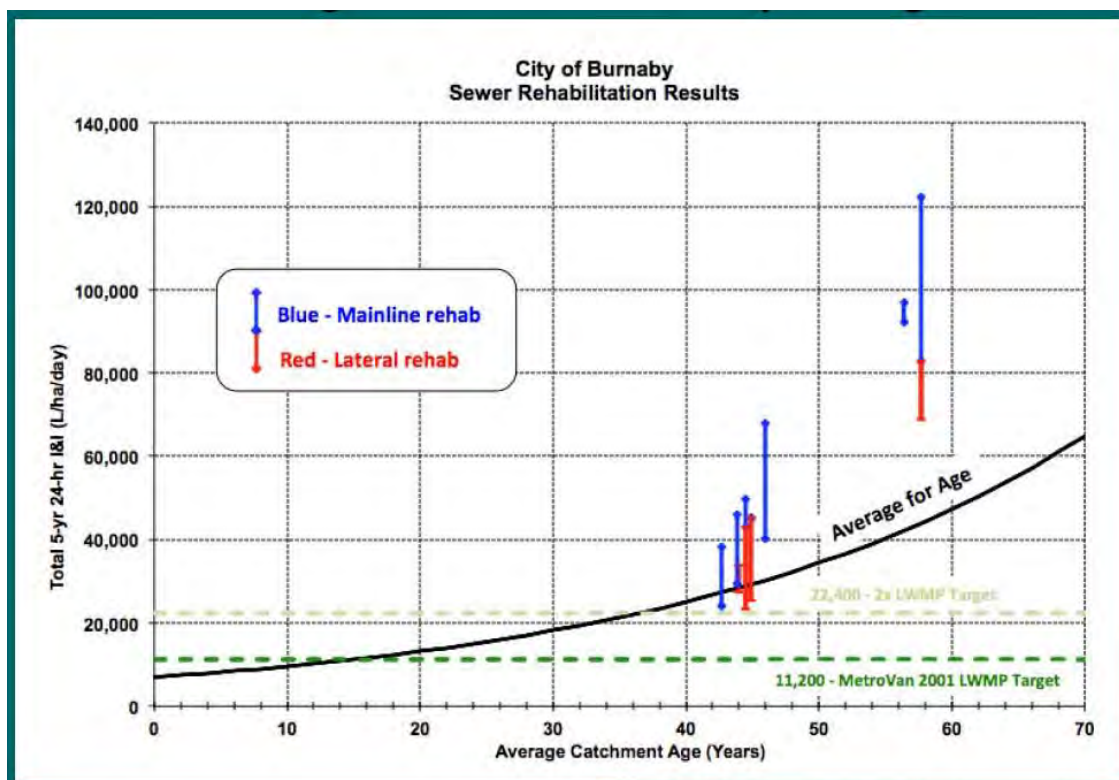


Figure 6 City of Burnaby I&I Reduction¹⁰

Based on I&I estimates in the Biennial Report: 2010-2012 (Liquid Waste Services Department, 2013a, pp. A8-A14), excessive I&I has been identified in the North Shore Sewerage Area from areas east of the Capilano River and in the Fraser Sewerage Area from areas north and east of New Westminster and from areas south of the Fraser River east of Highway 99 (Figure 7 and Figure 8). Groundwater inputs from construction dewatering, leachate discharges from landfills and continuous industrial discharges can mimic I&I. The affect that these inputs have on the Metro Vancouver I&I estimates has not been assessed.

¹⁰ Sewer lateral rehabilitation is only within the municipal right-of-way.

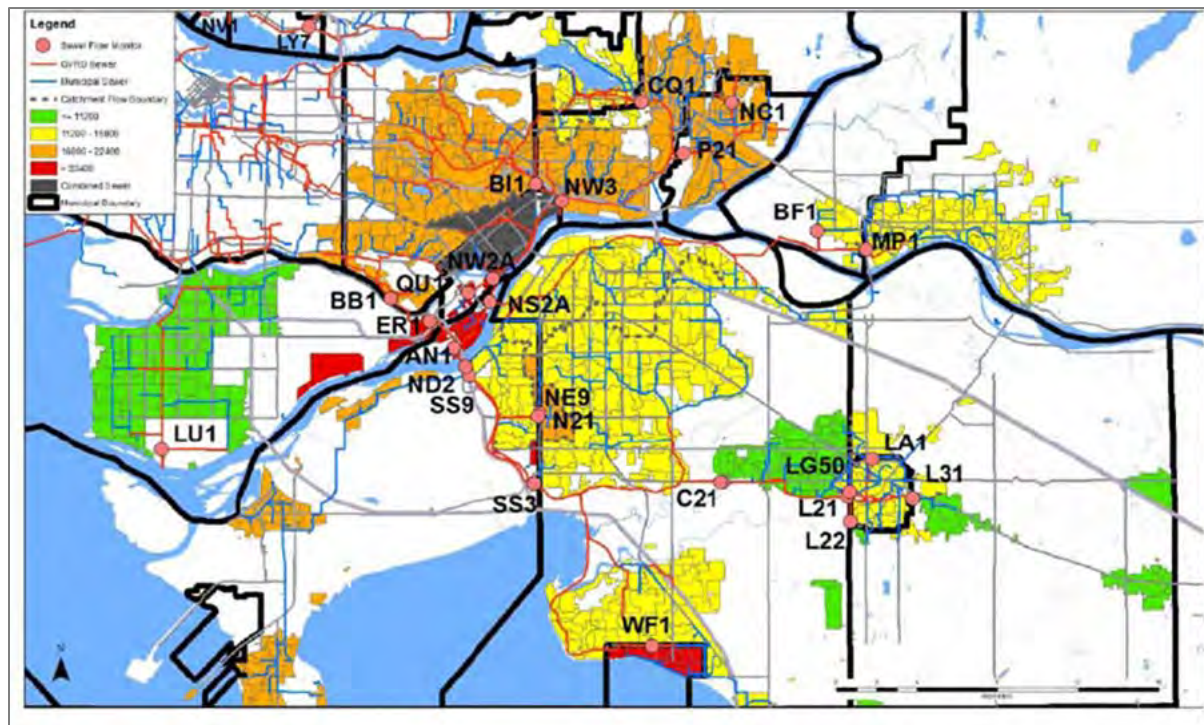


Figure 7 Estimated Fraser Sewerage Area I&I 2011-2012

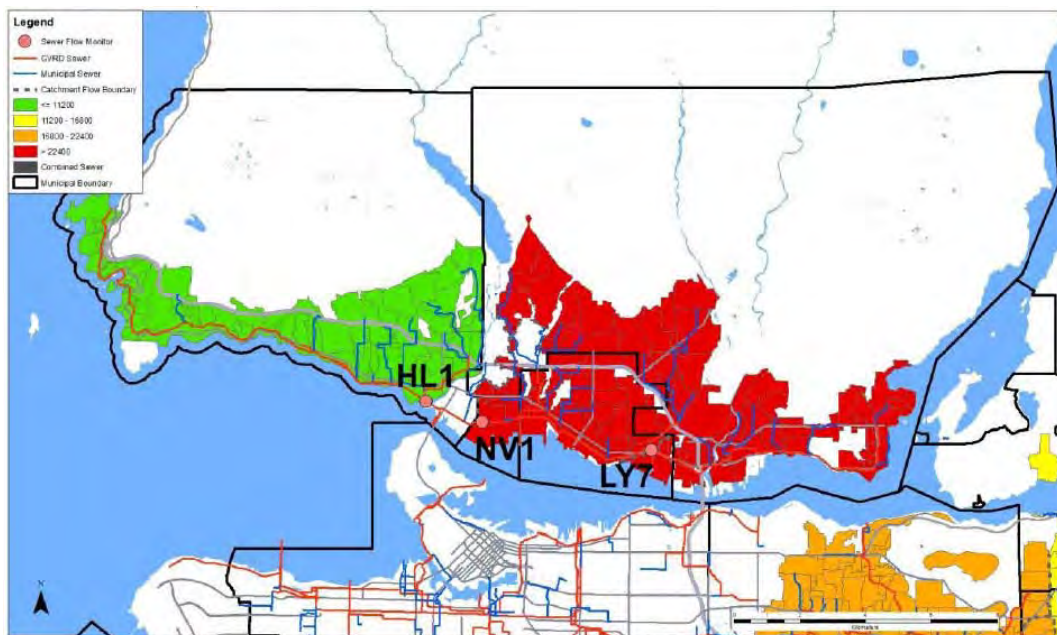


Figure 8 Estimates North Shore Sewerage Area I&I 2011-2012

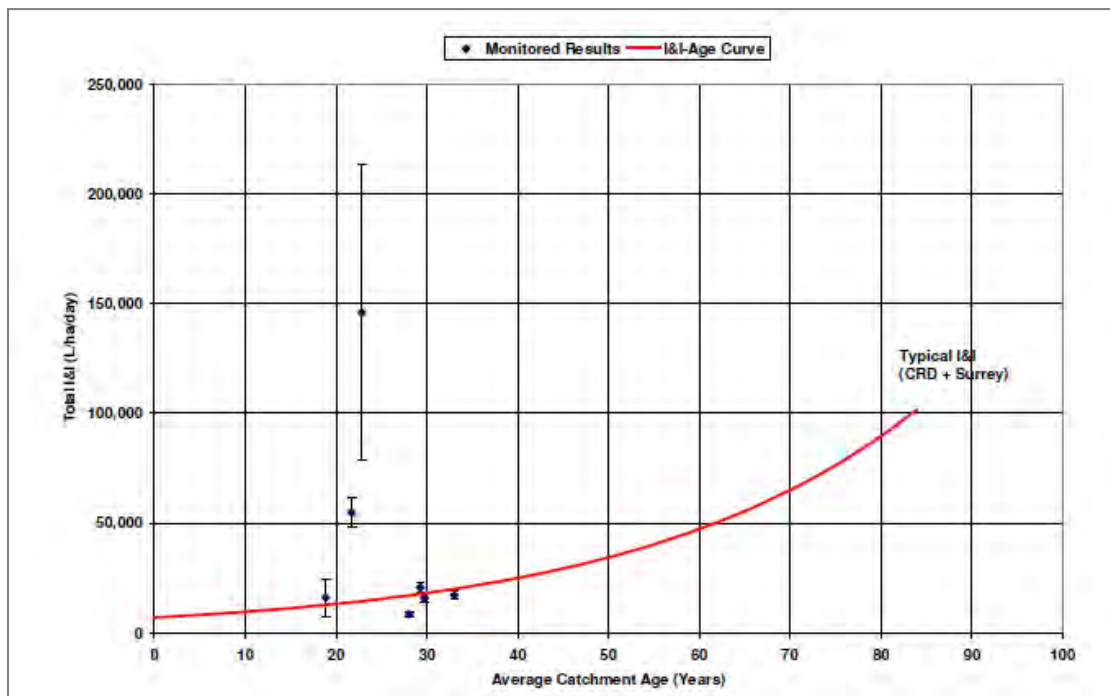


Figure 9 Relationship of Sewer Age to I&I Rate

Sewer Age and I&I Rates

In exploring I&I relationships in British Columbia, a correlation between sanitary sewer age and I&I rates is observed and shown in Figure 9 (Kerr Wood Leidal Assoc., 2011). Although this relationship is based on observations using the Envelope Method, it is not entirely clear which mechanisms are responsible for the relationship. Pipe materials and in-situ conditions are certainly factors in this relationship as well as quality of workmanship associated with the installation time. For sewers constructed prior to the 1970s (45 years or older), the majority were combined sewers or sanitary sewers without wastewater treatment. Additionally, semi-combined sewers were constructed in the Copley and Collingwood catchments of Vancouver in the 1940s and 1950s (Grill, 2014). Poor construction quality and cross-connections appear to have been issues starting with the region's first sanitary sewers (Blackman, 1917), with the main source of I&I being identified as poorly constructed new private sanitary laterals¹¹. As the sewer system expanded, it is not clear what sewer and lateral construction standards were enforced regarding gaskets in the pipe joints and the prohibition of cross-connection, particularly foundations.

¹¹New Westminster constructed separate sanitary and storm sewers until 1912 when combined sewers were allowed. Foundation drains were required to connect to storm sewers (Blackman, 1917).

Although separated sewer service laterals were required at the beginning of the 20th Century¹², this practice may not have been universal and appears to have changed by mid-century as evidenced by the number of foundation drains from pre-existing homes that were tied to sanitary sewers in Vancouver's Still Creek catchment during the 1940's and 50's. (Grill, 2014).

When comparing estimated I&I rates in the region, I&I rates tend to be greater for catchments with older, pre-1970s sewers (Liquid Waste Services Department, 2013b). It is possible that Figure 9 better represents a relationship between I&I and era of construction practices rather than pipe age.

I&I Management in Other Jurisdictions

The different approaches used for I&I design allowances by example communities in Australia-Asia, Canada, Europe and the United States are summarized in Appendix B: I&I Management in Other Jurisdictions. Approaches and allowances vary significantly from jurisdiction to jurisdiction. Consequently, there is not a single I&I allowance number or approach to I&I allowance that can be identified due in part to differences in sewer system design, local regulations and climate.

Semi-combined Sewers versus Sanitary Sewers

A key finding is that there appears to be two parallel yet fundamentally different types of sanitary sewers built: a true sanitary sewer without intended cross-connections, and a semi-combined sewer with building foundation drains intentionally connected to the sanitary sewer lateral.

While sewer systems in the jurisdictions studied are either sanitary or semi-combined sewer, the literature is not always clear as to which system is in use or to which system the I&I allowances are applied to. For example, in Ontario, semi-combined sewers are referred to as sanitary sewers on the basis that municipal stormwater is not conveyed in these systems even though building foundation drainage is conveyed. Drainage from roofs may add additional infiltration to the foundation drains. In these instances, storm sewers are typically shallow and collect primarily street runoff. A summary of Canadian I&I design allowances for semi-combined sewer systems is shown in Appendix B: I&I Management in Other Jurisdictions, Table 8.

In a 2006 survey of predominantly US municipalities, the physical condition of the private lateral, followed by cross-connections of foundation drains and roof downspouts were identified as the two most common I&I management issues (Strand Associates Inc., 2006, pp. A-1, A-2). It is unclear from this survey whether the issue of foundation drain cross-connections is a result of previously acceptable servicing practices or poor construction practices. Also, no distinction is made between sanitary sewer and semi-combined sewer.

As I&I reduction programs prioritize catchments with high I&I and the distinction between sanitary and semi-combined sewer is seldom made, it may be that a significant portion of the sanitary sewer I&I

¹² Based upon New Westminster's sewer system prior to 1912, storm sewers were constructed to be deep enough to drain residential basements and precluded the need for semi-combined sewers (Blackman, 1917).

mitigation work reported in literature is instead semi-combined sewer separation. This muddies our understanding of what is achievable and appropriate for sanitary sewer systems and introduces the question of how to best identify and manage semi-combined sewers, or systems which behave like them.

Approaches to I&I allowances in other jurisdictions do not provide clarity. The identification of semi-combined sewers in East Vancouver (Grill, 2014) and the high RDI&I rates associated with such sewer systems suggests that there are likely other catchments within Metro Vancouver that can be characterized as semi-combined sewers. If so, then it is unlikely a 11,200 L/ha-d I&I allowance can be reached in those catchments without undertaking a very comprehensive foundation drain cross-connection elimination program.

I&I Allowance Reassessment

I&I Scenarios

Metro Vancouver evaluated two scenarios for the regional sanitary sewer system where the timing of upgrades required to meet Metro Vancouver's basic service criterion¹³ were compared using different I&I allowances: 11,200 L/ha·d and 22,400 L/ha·d.

The findings are included in Appendix C: 2013 Estimated Financial Implications of Doubling the Inflow and Infiltration Allowance. The rationale behind this analysis is that accommodating an increased I&I allowance requires the use of growth capacity and would trigger growth capacity upgrades sooner. The undiscounted impact to the regional wastewater collection system is estimated to be \$475 million for the Fraser Sewerage Area and \$89 million for the Lulu Island West Sewerage Area. For the North Shore Sewerage Area, only the Hollyburn Interceptor would require upgrading at an estimated cost of \$35 million. Costs for capacity upgrades to the Lions Gate, Lulu Island and Northwest Langley Wastewater Treatment Plants were not factored into the analysis.

Accelerated WWTP capacity upgrades to the Annacis Island WWTP are estimated to be on the order of \$500 million as accommodating the additional I&I would trigger the next stage of planned upgrades.

The cost to provide the additional capacity to convey and treat an additional 11,200 L/ha·d of I&I for all three sewerage areas is on the order of \$1.1 billion as shown in Table 1.

Table 1: Comparison of Costs to Reduce I&I to 11,200 L/ha·d versus Convey and Treat 22,400 L/ha·d of I&I

Sewerage Area	Service Area (ha)	Additional Peak Flow for an Added 11,200 L/ha·d of I&I	Reduced Excessive I&I with Lateral Repairs ¹⁴		Cost of Capacity Upgrades to Convey and Treat Excessive I&I
			Potential Sites 0.5 L/s each site	Repair Costs \$6500/site	
Fraser (Annacis)	45,126	5.85 m ³ /s	11,700	\$76,050,000	\$975,000,000
Lulu Island West	4,936	0.64 m ³ /s	1,300	\$8,450,000	\$89,000,000
North Shore	7,491	0.97 m ³ /s	1,900	\$12,350,000	\$35,000,000
Total	—	—	14,900	\$96,850,000	\$1,099,000,000

Volumetrically, the additional 11,200 L/ha·d of I&I represents 5.5 m³/s of wet weather flow for the Fraser Sewerage Area, or approximately 50% of the current Annacis Island Wastewater Treatment Plant

¹³ Basic service is defined as peak dry weather flow plus inflow and infiltration allowance through interpretation of Policy 9, (Policy and Planning Department, 2002, p. 29)

¹⁴ The cost of lateral inspection and repair is highly variable. Preliminary observation from the District of North Vancouver indicates that approximately 0.5/s in inflow can be eliminated from cross-connection repair at a cost of \$6500 per connection (Carroll, 2014, p. 7).

capacity¹⁵. WWTP plant bypasses which do not meet effluent quality criteria are prohibited under the WSER; doubling the I&I allowance would require the equivalent of half the current Annacis Island plant capacity to treat occasional peak wet weather I&I flows.

Regional I&I Experience

The 11,200 L/ha·d I&I allowance used by Metro Vancouver to assess wastewater infrastructure provides a wet weather flow allowance approximately equal to peak dry weather flow. Excessive I&I is the flow in excess of the 11,200 L/ha·d allowance and contributes to reoccurring SSOs in the North Shore and Fraser Sewerage Areas.

Apart from SSOs, excessive I&I reduces the service life of existing sewers through loss of capacity and hastens growth upgrades. If SSOs due to excessive I&I are mitigated through regional accelerated capacity improvements or regional containment facilities, policies to apportion their respective costs should be reviewed to determine if such works place an unequal burden on municipalities with long-standing, effective I&I management programs as opposed to those from which the excessive I&I originates.

Causes of Excessive I&I

Except for areas of Burnaby, New Westminster, North Vancouver, Vancouver and the University Endowment Lands, the municipal sewer network was installed starting in the late 1950s. Much of this early system consisted of combined sewers, and prior to the early 1960s, lacked wastewater treatment.

Even after completion of the two original regional wastewater treatment plants at Iona Island and Lions Gate, many areas were serviced either by small first generation wastewater treatment plants (for example, located in Aldergrove, Central Surrey, Ladner, Central Richmond and White Rock) or discharged untreated wastewater via outfalls located at Burrard Inlet and the Fraser River.

It is believed that the current I&I situation has evolved from a combination of factors: material degradation over the decades; the sewer construction practices that were common between the 1950s and the mid-1970s; and do-it-yourself repairs and drainage modifications. The basis being the following:

- the construction materials used during this period may have since deteriorated;
- sanitary sewers were an exception in the 1950s and 1960s as most of the regional population was serviced by combined sewers;
- untreated dry weather sewage discharges to the Fraser River and Burrard Inlet continued until the early 1970s: additional groundwater and stormwater aided in their dilution;
- trades people at the time would have developed work habits based on constructing combined sewer systems or experience in other jurisdictions
- the absence of storm sewers or presence of ditches too shallow to drain basements led to foundations drains being connected to sanitary sewers;

¹⁵ Based on a representative maximum treated flow of approximately 11.2 m³/s on October 31, 2012.

- the level of quality control and inspection during construction may not have provided sufficient oversight, focussed on cross-connection prevention, nor tested for tight sanitary sewer pipes;
- over time, field tie-ins and disconnections introduce potential points for infiltration; and
- sufficient time has elapsed to allow for cross-connections to be created by 'do-it-yourself' home drainage and sewer repairs.

While the impacts of I&I on sewer capacity were known and documented by the 1930s (Lai, 2008), it is not known how this was applied to new construction, particularly private sewer laterals prior to the 1970s.

Affects of Combined Sewers

Within the region, portions of the Cities of Burnaby, New Westminster and Vancouver, and the University Endowment Lands still have combined sewers which will be separated as part of their long-term, ongoing actions under the LWMP¹⁶. While combined sewers in Burnaby and New Westminster introduce significant quantities of rainwater and groundwater into regional sanitary trunk sewers, the impacts from combined sewers should not be a distraction from the impacts that excessive I&I has on the loss of growth capacity and on SSOs. Under the LWMP, the Ministry of Environment permits the overflow of combined sewers during wet weather via a legacy outfall—it does not permit the overflow of sanitary sewer overflows for storm events less than 1:5 year return frequency¹⁷.

Municipal I&I Experience

Variation in I&I Rates

Figure 10 is a composite of the most recent I&I estimates provided by municipalities for their catchments. In managing I&I, municipalities have reported both success and difficulties in reducing I&I to the 11,200 L/ha-d level (Boyland & Weismiller, 2011; Cheung, 2012).

Factors which appear to influence the success of municipal programs include:

- the characteristics of the original sewer servicing with respect to materials and workmanship;
- inclusion or exclusion of private laterals from the I&I management program; and
- success in identifying and rectifying both municipal main and private lateral cross-connections.

Groundwater and Foundations

Although it might appear that high groundwater levels correlate with high I&I rates, high groundwater does not prevent sewer systems from being below the I&I targets as observed for Richmond (Bie, 2013) and Ladner (Cheung, 2012). For these locations, long standing quality control procedures for new sewer construction, including private and public laterals, and the absence of basements may be responsible in preventing excessive GDI&I.

¹⁶ Action 1.2.6 (Metro Vancouver, 2010, p. 17)

¹⁷ Actions 1.2.4 and 1.2.5 (Metro Vancouver, 2010, p. 17)

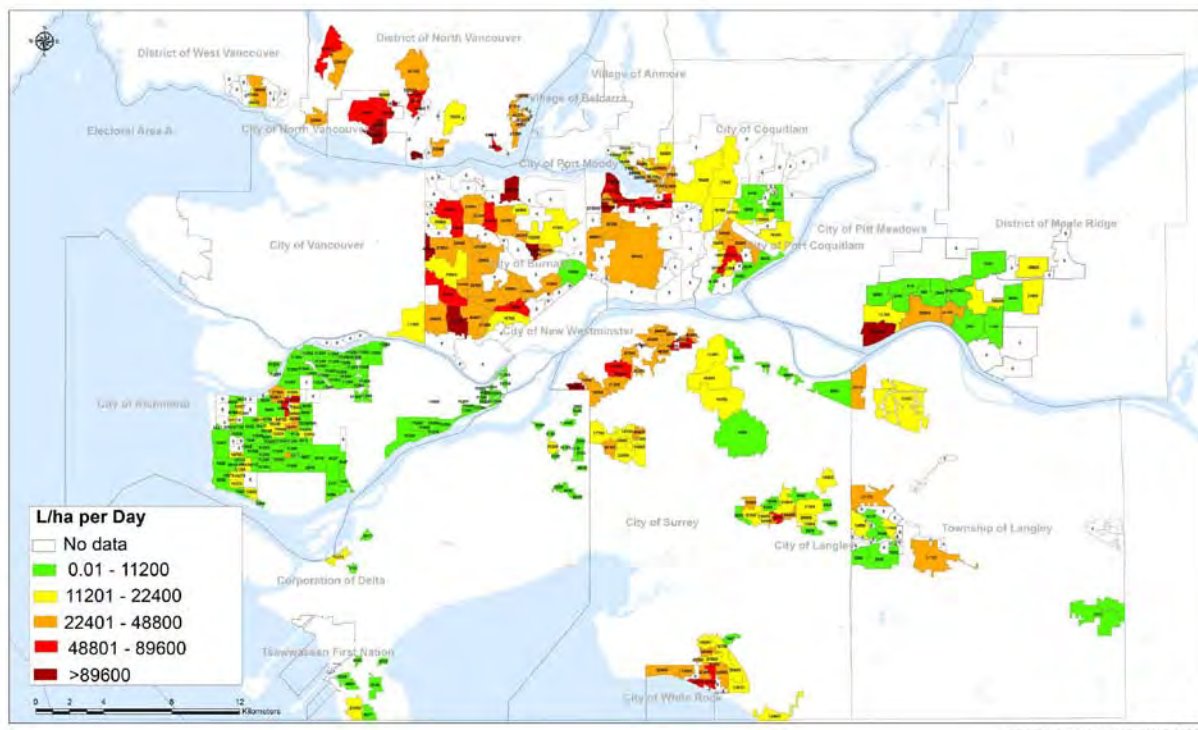


Figure 10 Reported I&I for Municipal Catchments

An assessment by City of North Vancouver in 2012 determined that GDI&I accounted for between 10% and 16% of the total estimated I&I (City of North Vancouver, 2012), ranging from 2,425 L/ha-d to 17,000 L/ha-d. In contrast, these catchments were found to experience significant RDI&I which accounted for 84% to 90% of the total I&I¹⁸. Based on the experience of Delta, the City of North Vancouver and Richmond, it is therefore possible that GDI&I may not be the priority component in I&I management.

However, the definition for GDI&I (as shown in Figure 4, page 9) best reflects slow groundwater infiltration with limited, if any, response to rainfall and may not adequately describe all groundwater infiltration processes.

For example, I&I monitoring results for semi-combined sewer catchments for properties with disconnected roof-leaders in Burlington, Ontario showed an immediate RDI&I response to rainfall (Štetić, 2009). In this study, foundation drainage connected to sanitary sewers responded as RDI&I and not as GDI&I. Similarly, in a foundation drain disconnection pilot study in London, Ontario, results indicate significant RDI&I removal upon the disconnection of foundation weeping-tiles from the sanitary sewers to the point that the sanitary sewer showed negligible response to rainfall (Chambers, 2013).

¹⁸ GDI&I was estimated to range from 2,425 L/ha-d to 17,000 L/ha-d, with RDI&I ranging from 12,570 L/ha-d to 132,000 L/ha-d, or 84% to 90% of total I&I (City of North Vancouver, 2012, p. 24),

Foundation drain connections to sanitary sewers were permitted during the 1940s and 1950s in Vancouver to both drain basements with inverts lower than nearby drainage ditches and to facilitate septic field elimination (Grill, 2014). The extent that this practice was permitted for similar situations in other GVS&DD municipalities is unknown.

When reviewing municipal catchments with I&I rates below 11,200 L/ha-d (Figure 10, page 22), generalized common characteristics include post-1970s construction and high groundwater tables which prevent the construction of basements. Conversely, common factors for areas with high I&I include pre-1970s construction, properties once served by onsite septic systems, and properties with or previously with ditched drainage.

Therefore, the importance of historical (pre-1970s) foundation drain cross-connection and the associated rapid hydrologic response to rainfall should be considered as a possible source of significant RDI&I.

Rainfall and Other Factors

There is wide variation in the Metro Vancouver region's rainfall. While rainfall influences how much RDI&I occurs, observed I&I rates among members shows that rainfall amounts do not determine a sewer system's I&I rate. For example, Metro Vancouver's I&I estimates for West Vancouver are significantly lower than the I&I estimates for White Rock, which receives less rain than West Vancouver (Liquid Waste Services Department, 2013a, pp. 10, 13)¹⁹. Municipal catchment data²⁰ also support this observation as illustrated by Maple Ridge's catchments with I&I rates below 11,200 L/ha-d when compared to catchments of Coquitlam or Burnaby where I&I rates are estimated to be greater than 33,000 L/ha-d (Liquid Waste Services Department, 2013b).

While precipitation and groundwater are factors which can contribute to excessive I&I, they do not necessitate high I&I rates or preclude the possibility of reducing excessive I&I.

Therefore, it is more likely that other factors such as the construction era of the sewers, sewer materials, construction and repair quality control procedures, in-situ environment and the prevalence of "do-it-yourself" repairs have resulted in a situation where municipalities are at different points with respect to their understanding and managing of I&I.

Municipal I&I Management Action

City of Burnaby

Ongoing I&I remediation work by Burnaby has focussed on Tier 1 I&I mitigation²¹. Results from Tier 1 mitigation indicate that I&I rates can be reduced to around the 25,000 L/ha-d in many cases. Recent

¹⁹ See Figure 7, page 15 and Figure 8, page 15 to compare I&I rates.

²⁰ As shown by Figure 10, page 22

²¹ Tiers are defined through the I&I management plan template (Kerr Wood Leidal Assoc., 2011) with Tier 1 being mainline sewers and lateral connections, Tier 2 being laterals in public rights-of-way, and Tier 3 being private laterals.

results are summarized by Table 2 (Boyland & Weismiller, 2011).

Comparing I&I rates with the age of sewers (Liquid Waste Services Department, 2013b, p. B12), data indicate that the high I&I catchments of Brentwood and Copley have pre-1960 sewers, whereas the lowest I&I is reported for catchments with post-1960 sewers: Forglen, Newcombe and Sperling.

As private laterals were not rehabilitated as part of this work, it is evident that upwards of 50% of a catchment's I&I may be attributed to I&I from private laterals.

Table 2 Burnaby Tier 1 I&I Reduction Results

Catchment Name	Initial I&I (L/ha·d)	Mainline Sewer (L/ha·d)	Lateral-Mainline Interface (L/ha·d)
Forglen	49,700	43,100	23,400
Royal Oak	44,000	45,300	25,500
Newcombe	33,800	incl.	27,500
Brentwood	122,300	82,800	68,800
Malvern	46,000	29,600	—
Sperling	38,100	23,890	—
Copley	96,900	92,200	—
South-Slope 2	67,800	40,400	—

City of Coquitlam

I&I management strategies presented by Coquitlam, Appendix D: City of Coquitlam: Inflow and Infiltration Management Strategy, indicates that the catchment with the greatest I&I is also the oldest with respect to the age of its sewers and service laterals.

One concern is that for many catchments I&I rates were extrapolated from limited flow monitoring that captured only small rainfall events due to timing and therefore, may be overestimated.

There also appears to be correlation in the occurrence of cross-connections in catchments where storm sewers were constructed following the elimination of drainage ditches in the 1990s. There is no mention if this is due to foundation drains having been connected to sanitary sewers due to the prior absence of storm sewers or to insufficient drainage gradient.

City of North Vancouver

The City of North Vancouver produced an I&I management strategy in 2012. The strategy indicates that some City sanitary sewers experience I&I rates on the order of 130,000 L/ha·d. About 60% of the City's sanitary sewer system is between 150 mm to 200 mm in diameter. The majority of the collection system (86%) is vitrified clay pipe with PVC pipe used in the most recent sections (about 13%) (City of North Vancouver, 2012).

Over 80% of the City's sewers were installed prior to 1970, with the oldest sewers dating back to the 1920s. All of the catchments identified as having I&I in excess of 67,100 L/ha·d were constructed prior to the 1950s. As well, pre-1950s era sewer construction correlate with the areas exhibiting high I&I rates – the oldest catchments having greatest I&I rates of around 130,000 L/ha·d.

The majority of the excessive I&I is RDI&I related which suggests significant cross-connections or very rapid groundwater infiltration in the problem catchments. However, mainline rehabilitation in a portion of the 1st and St David catchment, currently the worst I&I catchment in the City, yielded about 20% reduction in total I&I over two years of monitoring. In that area, only a few private-side storm drainage cross-connections were identified using smoke testing (Matsubara, 2014).

The City continues to investigate the catchments with excessively high I&I with the objective of finding mitigation options.

District of North Vancouver

Concentrated I&I reduction work by the District of North Vancouver was started in 2011 for the Lynn Valley area. Regulatory and financial drivers led to the initiation of an \$8 million program over five years.

Financial drivers for the targeted I&I reduction work include avoiding costly I&I storage needed to prevent reoccurring SSOs and avoiding costly sewer capacity upgrades needed for planned growth.

Cross-connection removal on private sewer laterals by the District so far shows significant promise as a positive cost/benefit technique to reduce inflow. This will inform a cost-benefit analysis which can be used to determine which I&I management components will be expanded to the broader District and as well as potentially being adopted by other member municipalities.

Early indications are that addressing inflow will be more economical on a dollar per litre of I&I basis as compared to addressing infiltration. It may be necessary that reduction in infiltration will be needed to reach the peak flow reduction target in the Lynn Valley area. The cost-benefit data produced from the District's I&I reduction program will continue to be developed and could become a reliable tool for other I&I programs.

An overview of the District's I&I reduction program is provided in Appendix E: District of North Vancouver: Inflow & Infiltration Reduction Program 2014 Status Report

City of Surrey

As part of their I&IMP, Surrey completed a volumetric assessment of the data collected from 64 temporary flow monitoring sites as well as available data from their sanitary lift stations and permanent flow monitoring locations. For example, I&I rates for three areas of regular study were estimated at 22,250 L/ha·d for North Surrey, 17,100 L/ha·d for Clayton-Cloverdale, and 17,900 L/ha·d for South Surrey.

Surrey has several problem areas with elevated I&I levels that have been the focus of sewer evaluation and rehabilitation programs to date. The City's approach to I&I evaluation and rehabilitation projects typically includes CCTV inspection, manhole inspection, smoke testing, pressure testing and grouting. The projects range from Tier 1 to Tier 2 and post rehabilitation data show that the City has achieved varying degrees of success in reducing I&I. Private laterals, Tier 3, were not part of the work²².

Surrey reports investing nearly \$5 million over the last few years with limited measureable improvements in I&I rates. Surrey's perspective based on their analysis is that the optimum benefit-to-cost is at Tier 1 and quickly diminishes towards Tiers 2 and 3.

City of Vancouver

Much of the area east of Nanaimo Street and south of First Avenue was built as separated sanitary sewer system. Starting in the 1990s, Vancouver has focussed on extensive cross-connection elimination work to improve the health of Still Creek. The City has determined that sanitary sewer servicing provided to many existing, formerly rural homes in the 1940s and 1950s had created a significant I&I problem in the City's sanitary system and Metro Vancouver's Copley and Collingwood sanitary catchments. This occurred as a result of connecting existing septic systems to new sanitary sewers and the private-side connection of weeping tiles to sanitary sewers laterals when local drainage ditches, which at one time lined many of the local streets, were too shallow to receive foundation drainage (Grill, 2014).

In addition to the legacy issues, cross-connections resulting from unpermitted, do-it-yourself plumbing modifications were found to be numerous. Through a combination of enforcement of the City's Plumbing Bylaw, where property owners are ordered to remove discovered cross-connections, and redevelopment, cross-connections and I&I rates have been significantly reduced.

District of West Vancouver

Work undertaken by the District of West Vancouver has reduced I&I rates to approach 11,200 L/ha·d as measured in the Hollyburn Interceptor (Liquid Waste Services Department, 2013b, pp. B355-B360)²³. District of West Vancouver public-side I&I programs focusing on the Ambleside area are credited with reducing the observed Metro Vancouver trunk sewer I&I rates. However, tributary catchment specific I&I rates reported by West Vancouver are between 13,000 and 26,000 L/ha·d, which is higher than the regional target. This discrepancy may be due to regional catchments reflecting attenuated I&I from cumulative municipal catchments and/or differences in municipal I&I rates projected for the 24h 1:5 year return storms (municipal) versus the measured regional I&I rates for a given year.

An overview of West Vancouver's I&I reduction program is provided in Appendix F: District of West Vancouver: Sewer Rehabilitation and Flow Monitoring Program.

²² *ibid*

²³ As the Metro Vancouver estimate for West Vancouver I&I rate decreased from greater than 22,400 L/ha·d to less 11,200 L/ha·d as associated with a <1:2 year storm event.

Issues and Challenges

I&I Estimates and Accuracy

Obtaining reliable sewer flow and precipitation data affordably is an underlying challenge for all I&I assessments. It is not always possible to capture the higher intensity storms during the monitoring period due to storm infrequency and the long-term costs of maintaining ongoing monitoring. Errors within monitoring equipment, variations in antecedent moisture conditions and localized rainfall variations also introduce some error to I&I estimates.

As no two storms will produce exactly the same I&I rates, even when the return frequency is estimated to be the same and when considering potential errors, I&I estimates are therefore inexact.

I&I and Era of Construction

A consistent observation among GVS&DD members' I&I programs is that higher I&I have been observed for catchments with pre-1970s sewer systems, specifically with the pre-1960s systems having much higher I&I rates than new systems. Extrapolating Burnaby's mitigation experience, a significant portion of the excessive I&I in older catchments would appear associated with private-side I&I. While municipal pipe inspection, repair and rehabilitation can focus on municipal mainline sewers with some success, pre-1970s private sewer laterals and cross-connections likely are contributing significant I&I in older catchments. Due to the number of serviced properties, the practicalities of inspecting and remediating private laterals and associated cross-connections are a major challenge.

Optimizing Mitigation & Private Laterals

While it is known that I&I from private lateral sewers can be a significant source of municipal I&I, and that addressing private lateral cross-connections and pipe condition are essential to successful I&I reduction programs (US Environmental Protection Agency, 1990; Strand Associates Inc., 2006; Herrera Environmental Consultants, 2009; Kerr Wood Leidal Assoc., 2011), local progress in this area is mixed (Liquid Waste Services Department, 2013b). One reason this may be the case, is a reliance on conventional benefit-to-cost analysis to direct limited municipal resources.

Certainly, developing inclusive benefit-to-cost analyses of I&I mitigation is made challenging by the uniqueness of sewer catchments and the two-tier accounting structure of municipal versus the regional sewerage systems. Analyses by Surrey and Seattle indicate that from solely a municipal perspective, Tier 1 I&I mitigation appears to provide the optimal benefits-to-costs (Burke, Jacobsen, Merrill, & Twenter, 2012, p. 22). However, when considering the whole system (municipal and regional), addressing private-lateral I&I through targeted programs is found to be the most cost-effective.²⁴

While member municipalities have the legal ability and can mandate private I&I management programs (Sheltair Group; Westcoast Environmental Law, 2008), such I&I programs have been limited to only targeted pilot studies to date. Metro Vancouver anticipates the work by the District of North Vancouver will provide better clarity on private lateral I&I mitigation and benefits-to-costs for such work.

²⁴ Appendix A: 1999 I&I Business Cases and Appendix C: 2013 Estimated Financial Implications of Doubling the Inflow and Infiltration Allowance

Translating I&I Allowances – Municipal and Regional

REAC-LWSC has enquired about a separate I&I allowance for municipal catchments, specifically a translation of the regional I&I allowance to address smaller sized municipal sewer catchments. Based on municipal experience, it has been suggested by some REAC-LWSC members that there may be an I&I reduction threshold at around 25,000 L/ha-d. In contrast, others members believe that the threshold to be lower, in part based upon observing municipal I&I rates close to or below the regional design allowance of 11,200 L/ha-d. In reviewing the local I&I estimates provided, I&I rates are still highly variable and no threshold appears obvious.

Municipalities were surveyed by Metro Vancouver in 2014 for their I&I allowances for new construction. These are shown in Table 3.

Table 3 GVS&DD Municipalities' I&I Design Allowances

Municipality	I&I Design Allowance (L/ha-d)	Municipality	I&I Design Allowance (L/ha-d)
Burnaby	11,200	Pitt Meadows	10,500
Coquitlam	11,200	Port Coquitlam	9,000
Delta	11,200	Port Moody	11,200
City of Langley	9,000	Richmond	11,200
Township of Langley	15,000	Surrey	11,200
Maple Ridge	11,200	Vancouver	11,200
New Westminster	14,688	UEL/UBC	Not reported
City of North Vancouver	8,640	West Vancouver	22,400
District of North Vancouver	16,800	White Rock	Not reported

Given the differences between the I&I allowances for new construction, as shown in Table 3, and I&I rates estimated from municipal flow monitoring, Metro Vancouver was asked to investigate if there may be merit in translating the regional I&I allowance to an allowance for municipal catchments. However, without detailed I&I monitoring for both the regional and tributary municipal catchments, it is only possible to speculate how appropriate adjustments could be undertaken.

One possible approach is to scale the regional I&I allowance. Comparison of the I&I estimated for the regional sewer catchments, Figure 7 and Figure 8, with municipal I&I estimates, Figure 10, shows I&I to be less in larger regional catchments than for their smaller, constituent municipal catchments.

The scalability of I&I rates to catchment size would depend on the hydrology of the sewer catchment, assuming that rainwater inflow and shallow infiltration would be the main sources of I&I—i.e., RDI&I. Under the same conditions, one might expect greater peak RDI&I rates for a small catchment compared to the RDI&I measured much further downstream in the larger catchment it is tributary to. On a per area basis, downstream peak flows would be expected to show some attenuation when compared to the upstream peak flow in the tributaries.

Attenuation effects are used in the design of the sanitary sewer component of wastewater where different peaking factors are used for large versus small catchments. However, peaking factors for sanitary sewers also represent diurnal sewer-use patterns and may be inappropriate to use outright as they are not purely indicative of hydrology. Even so, the ratios of population based peaking factors may help represent some of the peak I&I attenuation if it reflects some of the hydrologic characteristics of the time of concentration for rain derived inflow.

Factors which influence peak flow attenuation are the shape of a catchment, its slope and topography, size of sewers and number of pump stations. Thus, the role of attenuation in I&I cannot be generalized.

The Envelope Method attempts to overcome this by relating the total I&I estimate to a standardized 24 hour, 1:5 year return storm. However, for small catchments the translation to a 24 hour duration storm event may be problematic given that peak I&I flows are related to short times of concentration as compared to the larger catchments where peak flow would be determined by longer times of concentration.

Design Peaking Factors

In the design of sanitary sewers and combined sewers, peaking factors are applied to unit sanitary flow rates to represent peak magnitudes of sanitary flows. In sewer design, I&I allowances are typically added after the peak sanitary flows have been calculated. Only three exceptions²⁵ were identified with an I&I allowance included in the unit sanitary sewer design values: nationally in France, Canton Valais and the GVS&DD's original 1953 design criterion. In these instances, I&I design allowance is multiplied by the respective peaking factors for sanitary flow. The Harmon Peaking factor is one commonly used peaking factor for sanitary sewer design.

There do not appear to be any commonly used peaking factors to represent the hydrology of I&I and attenuation effects of larger catchments. Regardless, I&I attenuation does occur as flows from small tributary catchments travel downstream into the greater sewer system. A system specific hydrologic model could provide a characterization of how much attenuation is possible, with the objective of determining ratios or peaking factors of tributary to downstream I&I rates. However, this approach is anticipated to be time intensive and would generate catchment specific results that could not be extrapolated to other catchments.

Scalable I&I Rates

In the absence of the hydrologic characterization of the municipal and regional sanitary sewer network, estimated I&I for municipal catchments was scaled upwards based upon catchment population using the Harmon peak factors²⁶ to illustrate this concept. For comparison, adjusted and unadjusted maps use the same colour coding for catchment I&I performance, with the adjusted catchments allowing higher I&I rates for catchments with small populations. These are shown by Figure 11. Scaling was simplified as follows:

²⁵ See Table 8 Comparison of Residential Sewer Design Values, p 41 of this report.

²⁶ Per the Rawn Report, Figure 74 (Rawn, Hyde, & Oliver, 1953, p. 138)

- large catchments with a population of 30,000 or greater are unadjusted as the factor is $f=1$;
- medium catchments with a population less than 30,000 and greater than 10,000 are adjusted by a factor of $f=1.2$ based on $1.81/1.5$; and
- small catchments with a population less than 10,000 are adjusted by a factor of $f=1.5$ based on $2.25/1.5$.

The scaling assumes that at an I&I allowance of 11,200 L/ha·d as measured in Metro Vancouver's trunk sewers translates to a comparable I&I rate of 16,880 L/ha·d for a small catchment.

Furthermore, alternate population peaking factors lead to different scaling. For example, in very small catchments, an I&I allowance of 22,400 L/ha·d I&I would be representative of the up-scaled regional I&I allowance if based on the sewer flow peaking factors used in Brisbane (Queensland Urban Utilities, 2010). Identifying scaling factors appropriate to the Metro Vancouver region has not been undertaken.

As an alternative, to adjust I&I rates based on sewer catchment size and not population density, a relationship would need to be developed. At this time, it is not clear if the relationship would be any more meaningful than using population densities.

Prioritization

As municipalities work to reduce any excessive I&I, prioritization will be required to direct resources to the sewer catchments which can produce the most beneficial I&I reduction. This necessitates an understanding of I&I in the sewer system and require ongoing program commitments not just to the municipal mains but also to addressing excessive I&I from private lateral sources. Municipal records regarding the age of pipe, age of housing stock, rate of redevelopment and past ditch elimination programs may help to identify catchments which should be assessed for excessive I&I.

Once problem I&I catchments have been identified, subsequent prioritization is required for mitigation work. Within the I&IMP template, the approach to I&I prioritization is to focus on catchments with the highest I&I rates (Kerr Wood Leidal Assoc., 2011), as shown by Figure 12.

Catchments with excessively high I&I would receive priority mitigation. One further option to assist in the prioritization would be to implement a triage approach where catchments with I&I above certain thresholds would receive different levels of priority. For example, based on Figure 10, page 22, top priority would be mitigating catchments with I&I above 89,600 L/ha·d (eight times the I&I allowance). A secondary priority could focus on catchments with I&I above 44,800 L/ha·d, and so forth for other thresholds.

This approach appears appropriate to prioritize catchments which have exceedingly high I&I, for example greater than 89,600 L/ha·d, as these I&I rates would not significantly be reduced by applying I&I allowance scaling factors.

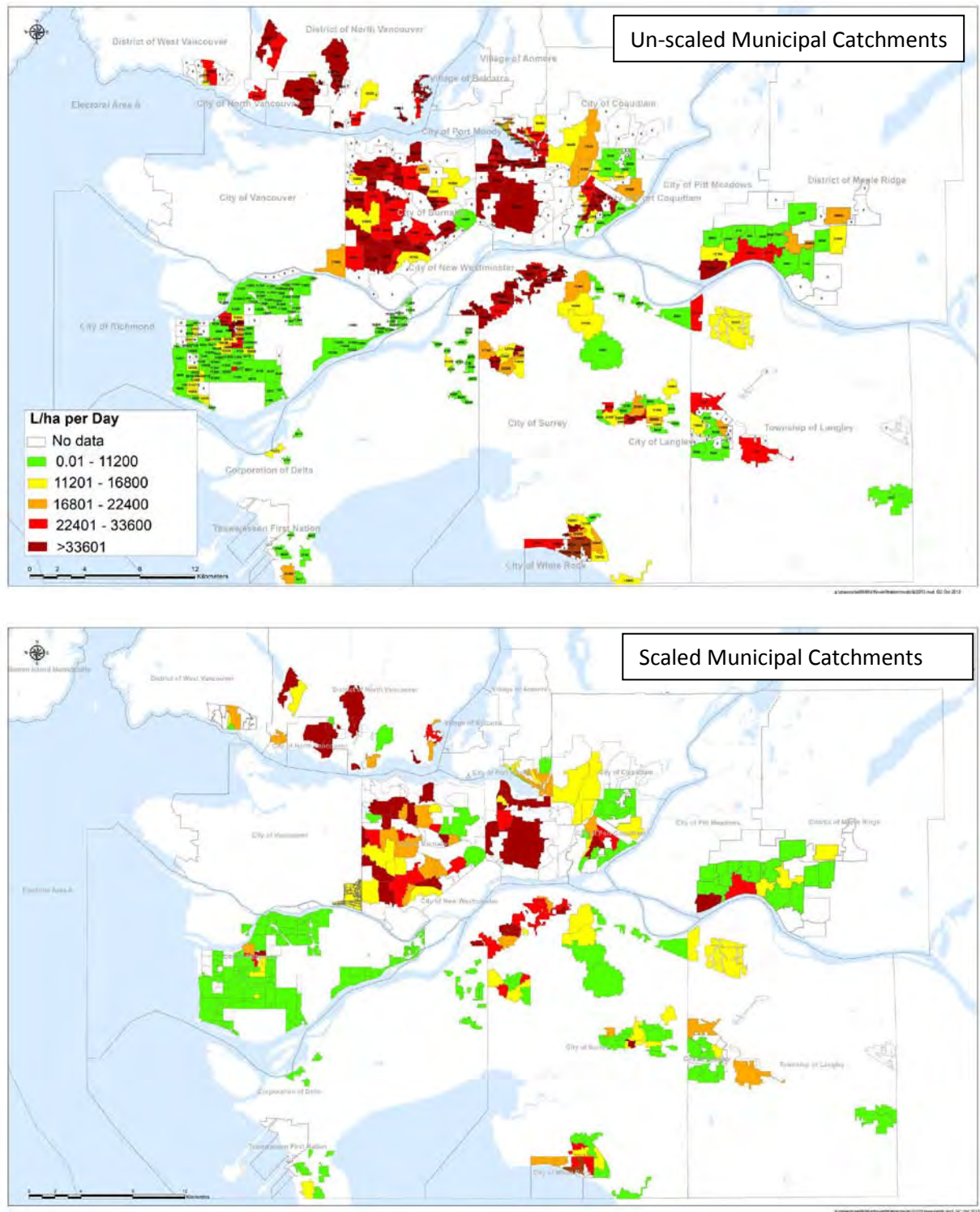


Figure 11 Comparison of I&I Reported with I&I Scaled by Municipal Catchment Population



Figure 12 I&I Management Triage

However, setting priorities just on I&I rates may be inefficient, particularly when I&I rates are in a lower triage level. Catchment size and the total I&I flow rates are considerations that also need to be included in the prioritization process. For example, a 100 ha catchment with I&I of 40,000 L/ha·d will contribute less I&I to the regional sewers than a 300 ha catchment with an I&I rate of 20,000 L/ha·d.

Therefore, municipalities need to consider both the peak I&I rates and total cumulative excessive I&I flow at the connection point to the GVS&DD sewers when prioritizing their I&I mitigation efforts.

Future Considerations

Climate change modelling for the British Columbia, South Coast region projects changes in seasonal precipitation: a 6% increase in Winter; a 7% increase in Spring; a 13% decrease in Summer; and a 9% increase in Autumn for 2050 (Climate Change Effects on Watershed Processes in British Columbia, 2010, p. 711). Additional factors, such as the PDO and ENSO, are anticipated to vary the magnitudes of these changes.

It is very likely that this will lead to increased rainfall intensity, which implies that a 1:10 year return storm event today will likely become a more frequent event in the future (BGC Engineering Inc., 2009a). As the regional I&I allowance is linked to the 1:5 year return storm event through the ILWRM, it is possible that a sewer system designed for an I&I allowance of 11,200 L/ha·d today would experience higher I&I rates when subjected to projected future rainfall intensities and patterns²⁷.

²⁷ through the ILWRM Actions 1.2.4 and 1.3.2 for SSO reoccurrence and basic service capacity (Metro Vancouver, 2010, pp. 17, 18)

Conclusions

Regional I&I Allowance

Metro Vancouver's financial analyses of increasing the I&I allowance are consistent and indicates that preventing excessive I&I is more cost effective than increasing conveyance and treatment capacity. However, the analyses did not examine whether there are any thresholds where I&I reductions are no longer feasible.

The current regional I&I allowance was introduced in the 1970s for GVS&DD sewer design based upon research done by Seattle. Similar I&I rates are still used by King County municipalities to design new sewers. However, Metro Vancouver's use of the 11,200 L/ha·d I&I allowance to estimate basic service flows and determine capacity upgrades does not recognize the current state of I&I in tributary municipal catchments and therefore is different than the I&I allowance's initial purpose – new sewer design. The basic service approach assumes that municipalities are actively reducing I&I to approximate the regional allowance and new sewer conditions.

For large catchment, I&I rates below or near the Metro Vancouver allowance of 11,200 L/ha·d are documented for several GVS&DD members (Liquid Waste Services Department, 2013a; Liquid Waste Services Department, 2013b). While attaining the I&I allowance target is technically achievable, based upon local municipal experience, there remains uncertainty as to whether there might be a threshold beyond which reductions diminish. It is unknown if such a threshold is influenced by the suspected presence of semi-combined sewers or by other factors.

Common characteristics among catchments with I&I near or below the regional I&I allowance includes: the absence of basements and deep foundation drains, the presence of storm sewers which dates from the same time as the sanitary sewers, and typically post-1970s construction. Conversely, common characteristics of catchments with excessive I&I include pre-1970s servicing, drainage initially or still provided by ditches, the presence of pre-1970s buildings with basements, and connection of former septic systems to sanitary sewers.

In evaluating I&I design allowances for other jurisdictions, there appears to be significant variations between sewer systems, I&I definitions and design methodologies. Consequently, I&I targets defined as acceptable in one jurisdiction do not necessarily translate as acceptable I&I in another. The extents to which foundation drainage is connected to sanitary sewers outside of British Columbia complicates the ability to compare I&I reduction programs and rates with the rest of Canada. In addition, the prevalence of semi-combined sewers and their reference as sanitary sewers by other jurisdictions complicates meaningful comparison of I&I rates.

The identification of mid-20th Century foundation drainage to early-20th Century sanitary sewers in Vancouver documents the construction of “semi-combined like” sewers in the region and indicates that,

for a time, it was accepted practice to allow building foundation drainage to connect to sanitary sewers. It is therefore conceivable that other pre-1970s municipal catchments can be characterized as semi-combined sewers. However, only detailed examination of municipal records and focussed I&I studies are likely able to determine the extents of semi-combined sewers.

Should a large numbers of semi-combined catchments be identified, then identifying distinct interim and long-term I&I allowances may be appropriate: based on experience in Toronto, 22,400 L/ha-d may be a suitable semi-combined sewer allowance, while 11,200 L/ha-d can remain the long-term allowance for all types of sewer systems.

Irrespective, additional research and evaluation of municipal sewer catchments is warranted to determine the extent of semi-combined sewers in the region.

Municipal I&I Allowance

Although it may seem plausible that I&I rates can be higher for small catchments and with peak I&I attenuating in larger catchments, identifying an equivalent I&I allowance for municipal sewer catchments was inconclusive.

Attempting to scale the Metro Vancouver I&I allowance for small municipal catchments is complicated by unique hydrologic characteristics for each sewer-shed and the absence of any similar methodologies being used elsewhere. However, the scaling of I&I results can help to differentiate areas with significant I&I concerns—akin to a sensitivity analysis. It is evident that irrespective of whether the I&I design allowance is 11,200 L/ha-d or 22,400 L/ha-d, excessive I&I must be reduced. Based on I&I design values from across Canada, it appears that all I&I rates above 25,000 L/ha-d can be regarded as excessive.

Considering experience from Seattle and elsewhere, addressing excessive inflow from private laterals must be a key component of municipal I&I reduction programs. Such programs will take time and require a long-term commitment with respect to funding and resources; their initiation and continuation cannot be deferred if I&I is to be controlled to minimize SSOs and ensure that conveyance capacity is available for community growth. The continuation and updating of municipal I&I programs will also be required to prevent new I&I problems and contain long-term sewerage infrastructure costs.

Ultimately, building stock renewal cycles and redevelopment will also help rectify drainage cross-connections and reduce inflow from private laterals. For the interim, targeted I&I programs are needed for catchments with extremely high I&I rates.

Recommendations

The need for I&I management is to ensure that sufficient sewer capacity is available for both current services and designed future service levels, thereby permitting community growth and minimizing the likelihood of sewer overflows. The ILWRM identifies municipal actions and responsibilities for I&I management planning and an I&I target rate. These include:

- A. Using the 2011 I&I Management Plan Template as a guide in developing I&IMPs
- B. Implement I&IMPs to ensure that I&I rates are within Metro Vancouver's I&I allowance at GVS&DD connection points.

In consideration of the assessment of the I&I allowance in this report, the following recommendations were made by the REAC-LWSC at their June 19, 2014 meeting:

1. That GVS&DD members use the "Envelope Method" as described in the 2011 *I&I Management Plan Template* to define I&I
2. That GVS&DD members share through Metro Vancouver the information they have learned implementing their I&I management plans to help ascertain if there are I&I reduction thresholds for old sewer catchments
3. That GVS&DD members work with Metro Vancouver to determine whether semi-combined sewers (sanitary sewers designed to accept foundation drain connections) are a significant factor affecting I&I rates and identify options to remedy semi-combined sewers
4. That, for the interim, Metro Vancouver continues to use 11,200 L/ha·d as the I&I allowance for basic service modeling for GVS&DD sanitary sewers and for capacity upgrades
5. That GVS&DD members will prioritize I&I management in sanitary sewer catchments tributary to reoccurring SSOs
6. That GVS&DD members will prioritize sanitary sewer catchments with the highest I&I rates for I&I reduction and mitigation
7. That this report and recommendations should be reviewed in four years

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Appendix A: 1999 I&I Business Cases

As part of the 1999 evaluation of options for managing Infiltration and inflow (I/I), studies were completed for the Fraser and North Shore Sewerage Areas that included analyses of the costs of variations approaches. The scenarios are summarized with costs expressed non-adjusted in 1999 dollars and use slightly different approaches and assumptions.

Fraser Sewerage and Lulu Island Sewerage Areas

In the Fraser Sewerage Area and Lulu Island Sewerage Area, the option with the lowest cost is a targeted rehabilitation/replacement of the conveyance system including GVS&DD, municipal and private pipes and connections as well as some WWTP upgrades. The rehabilitation is targeted on those areas with the greatest amount of rainfall-dependant infiltration and inflow (RDI&I) (as opposed to groundwater infiltration), followed by areas with lower RDI&I values. These scenarios include WWTP upgrades and GVS&DD collection system upgrades that maintain basic service with RDI&I values of 11,200 L/ha/d.

RDI&I of 11,200 L/ha/d is considered the allowable basic service amount while RDI&I of more than 22,400 L/ha/d is considered excessive.

For the Fraser Sewerage Area, the total discounted cost of the targeted RDI&I approach over a 75-year period using a 5% real discount rate is \$575 million in 1999 dollars. In comparison, the scenario where pipes are replaced based on age, the discounted cost is \$791 million in 1999 dollars.

For the Lulu Island Sewerage Area, the targeted RDI&I reduction cost is \$39 million in 1999 dollars while the scenario to replace pipes based on age costs \$85 million in 1999 dollars.

Assumptions for the Analyses

WWTP Scenarios

- A) *3xAAF*- Assumes that the ratio of average daily peak wet weather flow (PWWF) to average annual flow (AAF) at the plant linearly increases from 2.0 times AAF (current value) to 3.0 times AAF.
- B) *2xAAF* – The current plant liquid stream design for PWWF remains at 2.0 times AAF over time.
- C) *Basic Service* – The plant liquid stream design for PWWF at the plant is assumed to be equivalent to the Basic Service flow allowance of PDWF plus 11,200 L/ha·d .

GVS&DD Collection system Scenarios

- A) *Increased Conveyance (high)* – Assumes that increased levels of I/I will require future collection system upgrades to occur sooner to keep up with increasing wet weather flows.
- B) *Increased Conveyance (medium)* – Acceleration cost of upgrades assumed to be mid-way between those required for A) and C).
- C) *Basic Service* – Facilities are upgraded when flows exceed the Basic Service flow criteria of PDWF plus 11,200 L/ha·d.

Municipal System and Private Lateral Scenarios

- 1) *Age* – Replace municipal collection systems and private laterals at the end of their assumed life span of 100 years.
- 2) *Target Rainfall Dependent Infiltration/Inflow (RDI/I)* – Implement a targeted rehabilitation program aimed at rehabilitating the areas in the region with the highest RDI/I rates first.
- 3) *Target Groundwater Infiltration (GWI)* – Implement a targeted rehabilitation program aimed at rehabilitating the areas in the region with the highest GWI rates first.
- 4) *Target Total Infiltration* – Implement a targeted rehabilitation program aimed at rehabilitating the areas in the region with the highest total RDI/I and GWI. This scenario represents the highest effort between 2) and 3).
- 5) *0.5% Rehabilitation /Year* – Rehabilitate 0.5% of the system per year with priority based on age, I/I rates or some arbitrarily set criteria.
- 6) *1% Rehabilitation/Year* – Rehabilitate 1% of the system per year with priority based on age, I/I rates, or some arbitrarily set criteria.

Fraser Sewerage Area and Lulu Island West Sewerage Area RDI&I Scenarios

Table 4 shows the undiscounted and discounted costs of the modeled scenarios for Fraser Sewerage Area, while Table 5 shows the scenarios for the Lulu Island West Sewerage Area. These are coded based upon the above listed assumptions: (Letter of WWTP scenario) + (Letter of Collection scenario) + (Numeral of Municipal and lateral scenario). Scenarios CC2 and CC5 were identified as the least costly for the Fraser Sewerage Area, and

Table 4: Cost Summary for Representative Scenarios for Fraser Sewerage Area (Million Dollars 1999)

Scenario	WWTP	GVS&DD Collection System	Municipal	Private Laterals	Total
75-Year Capital Cost					
AA1	\$1,562	\$432	\$1,672	\$1,744	\$5,414
BB5	\$777	\$388	\$792	\$616	\$2,572
BB2	\$777	\$388	\$1,020	\$870	\$3,054
CC2	\$557	\$343	\$1,020	\$870	\$2,790
CC5	\$557	\$343	\$792	\$616	\$2,307
CC6	\$557	\$343	\$1,346	\$1,231	\$3,477
CC3	\$557	\$343	\$1,696	\$1,621	\$4,217
CC4	\$557	\$343	\$1,749	\$1,749	\$4,461
NPV @5%					
AA1	\$348	\$158	\$149	\$136	\$791

BB5	\$192	\$136	\$203	\$158	\$689
BB2	\$192	\$136	\$184	\$137	\$649
CC2	\$141	\$113	\$184	\$137	\$575
CC5	\$141	\$113	\$203	\$158	\$616
CC6	\$141	\$113	\$345	\$316	\$916
CC3	\$141	\$113	\$366	\$338	\$958
CC4	\$141	\$113	\$394	\$371	\$1,019

Table 5: Cost Summary for Representative Scenarios for Lulu Island Sewerage Area (Million Dollars 1999)

Scenario	WWTP	GVS&DD Collection System	Municipal	Private Laterals	Total
75-Year Capital Cost					
AA1	\$174	\$33	\$256	\$251	\$714
BB4	\$78	\$31	\$103	\$82	\$295
BB2	\$78	\$31	\$56	\$31	\$197
CC2	\$65	\$29	\$56	\$31	\$181
CC3	\$65	\$29	\$231	\$224	\$549
CC4	\$65	\$29	\$103	\$82	\$279
CC5	\$65	\$29	\$177	\$165	\$435
NPV @5%					
AA1	\$46	\$11	\$17	\$10	\$85
BB4	\$26	\$9	\$26	\$21	\$83
BB2	\$26	\$9	\$9	\$1	\$45
CC2	\$22	\$7	\$9	\$1	\$39
CC3	\$22	\$7	\$20	\$14	\$63
CC4	\$22	\$7	\$26	\$21	\$76
CC5	\$22	\$7	\$45	\$42	\$116

North Shore Sewerage Area

The North Shore Sewerage Area study assumes a 150 year timeline (it takes 150 years to replace the system assuming 0.67% per year replacement). This study uses a 6% real discount rate which will make costs that occur over 20 years in the future virtually zero in present value terms. Therefore, options that have costs delayed far in the future will tend to be favoured in present value terms.

Infiltration was assumed to be about 40% from in-street pipes, 5% manholes, 15% municipal connections, 25% private connections and 15% inflow.

As with the Fraser Sewerage Area and Lulu Island Sewerage Area evaluations, the North Shore Sewerage Area study evaluates a number of options that either emphasizes increasing conveyance and treatment capacity *or* rehabilitation and replacement of infrastructure to reduce RDI&I.

The options evaluated in the study are shown in Table 6 with their undiscounted and discounted costs.

Table 6: North Shore Sewerage Area Option Cost Summary

RDI&I Option	Capital Cost (Millions 1999 Dollars)					Total NPV @6%
	Conveyance and Treatment	Municipal system Replacement	Municipal and Private System Rehabilitation	Municipal Inflow	Total	
Level I-Base	9.4	735	0	0	745	74
IA-1	9.4	735	224	13	981	139
IA-2	9.4	735	109	13	866	116
IA-3	9.4	735	109	13	866	104
IB-1	9.4	735	90	4	838	106
Level II-Base	25	735	0	0	760	82
IIA-1	25	735	87	13	860	113
IIA-2	25	735	19	13	792	95
IIA-3	25	735	48	13	821	101
IIB-1	25	735	78	3	841	107
IIB-2	25	735	28	0.4	788	91
Level III-Base	43	735	0	0	778	91
IIIA-1	43	735	224	13	1015	156

The base level options have the lowest NPV but they also provide the smallest reductions in RDI&I.

In order to select the dominant options, the North Shore Sewerage Area study authors used a weighting system comprising NPV, RDI&I reduction, and overflow reduction. The weighting was varied as a sort of sensitivity analysis and the results are shown in Table 7.

Table 7: NSSA I&I Reduction Sensitivity Analysis

Three Weighting Factor Sets Each with Three Options of Lowest NPV		
NPV: 0.6 RDI&I: 0.3 Overflow: 0.1	NPV: 0.5 RDI&I: 0.4 Overflow: 0.1	NPV: 0.4 RDI&I: 0.5 Overflow: 0.1
Level I-Base	Level II B-2	Level I B-3
Level II B-2	Level I B-3	Level II B-2
Level I B-3	Level I A-3	Level I A-3

Reductions in RDI&I come at a cost and RDI&I itself has a cost in that without reductions in RDI&I, conveyance systems and treatment capacity will have to be increased, which can be costly. The idea should be to have the least cost wastewater system that satisfies the regulatory requirements.

Nevertheless, there are two options that are in the top three ranked for all three weighting structures: Level II B-2 and Level I B-3. The NPV of option II B-2 is \$91 million (1999\$) and for I B-3 it is about \$98 million (1999\$).²⁸ Both options target priority basins for rehabilitation as opposed to using a sewerage area-wide approach. This is similar to the options recommended for Fraser Sewerage Area and Lulu Island Sewerage Area which use a targeted rehabilitation approach to tackle the areas with the greatest amount of RDI&I. Focussing on areas with the greatest RDI&I was identified as the strategy because it provides the greatest reduction in RDI&I per dollar spent.

²⁸ Option I B-3 does not appear in Table 5 provided in the study but does appear in the figure that ranks the options based on NPV. Option I B-3 appears between Option II A-2 (NPV \$95 million) and Option II A-3 (NPV \$101 million).

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Appendix B: I&I Management in Other Jurisdictions

I&I design allowances and approaches to I&I regulation were reviewed for municipalities in Canada, United States, Australia, New Zealand, France, Switzerland, Germany, Norway, United Kingdom and Hong Kong. Due to the extents of combined sewers in some jurisdictions, I&I appears to be treated as a low priority by such jurisdictions.

In Europe, modern subdivision and urbanization projects require separated sewer systems, often connecting into combined systems. For these systems, the problems of excessive I&I are typically presented as added financial costs for pumping and treatment and less as a factor in triggering sanitary sewer overflows. I&I is often expressed as percentage of ADWF or AAF.

I&I Defined by Catchment Area

Cities in Australia, Canada, New Zealand and the United States approach I&I in a manner much more familiar and comparable to Metro Vancouver. Table 8 summarizes I&I design allowances for Canadian cities with sanitary sewers and with all or partial semi-combined sewers.

Table 8 I&I Design Allowance for Semi-Combined and Sanitary Sewer Systems

Municipality	I&I Design Allowance	Sanitary (S) or Partial Semi-Combined (SC)	Municipality	I&I Design Allowance	Sanitary (S) or Partial Semi-Combined (SC)
City of Calgary (Inflow and Infiltration Process Task Force, 2008)	24,192 L/ha·d	SC	Region of Peel (Inflow and Infiltration Process Task Force, 2008)	17,280 L/ha·d	SC
City of Chilliwack (Inflow and Infiltration Process Task Force, 2008)	8,640 L/ha·d + ADWF	S	City of Saskatoon (Inflow and Infiltration Process Task Force, 2008)	6,912 L/ha·d ²⁹	SC
City of Edmonton (City of Edmonton, 2013)	24,192 L/ha·d	SC	Greater Sudbury (Greater Sudbury, 2012, p. 17)	11,250 L/ha·d to 44,900 L/ha·d ³⁰	SC
Region of Halton (Inflow and Infiltration Process Task Force, 2008)	24,710 L/ha·d	SC	Region of York (Inflow and Infiltration Process Task Force, 2008)	17,280 L/ha·d	SC
City of London (Inflow and Infiltration Process Task Force, 2008)	8,640 L/ha·d	SC			

²⁹ There is an alternate allowance of 0.05 L/s per weeping tile connection.

³⁰ Depends upon the neighbourhood identified in Table 2.2

I&I Allowance and Return-Periods

Past I&I studies have compared I&I management programs from different jurisdictions in Canada and the United States. Although I&I measurement programs are similar, the quantification and definitions of I&I rates and corresponding targets vary from jurisdiction to jurisdiction. Care is needed in comparing experiences and I&I targets. For example, I&I target rates for King County are associated with storms with a 1:20 year return frequency (Wastewater Treatment Division, 2004, p. 2). Therefore, one needs to consider that different I&I allowances would be appropriate for a 1:20 year return frequency storm as compared for Metro Vancouver's 1:5 year return frequency storm.

Population Equivalent I&I and Peaking Factors

For smaller catchments, greater I&I allowance may be expected as the peaking factors of smaller sewer branches are greater than for trunk sewers. This is consistent with the Rawn Report's original design criteria which identified sewer peaking factors of 2.5 for a catchment population of 10,000, and 1.5 for a catchment population of 100,000 for sewer design (Rawn, Hyde, & Oliver, 1953, p. 138).

Sewer design based on equivalent population factors is an accepted practice with I&I allowances sometimes built into sewer design values. A comparison of values³¹ among jurisdictions is shown by Table 9.

Queensland represents I&I as a function of population and uses two different methods:

The first method uses wet weather peaking factors³² applied to a base sewer rate of 210 L/cap-d to develop the ADWF; the default for PWWF is given as five times ADWF for conventional sewer design.

The second method uses the "nu-sewer" design, which is PDWF plus an inflow allowance of 360 L/cap-d, where PDWF is defined as base sanitary flow of 150 L/cap-d multiplied by a peaking factor plus groundwater infiltration allowance of 30 L/cap-d.

In the case of the "nu-sewer" design, peaking factors are applied to only the base sanitary flow and not the inflow or infiltration allowances (Queensland Urban Utilities, 2010, p. 20).

Peaking factors added to peak dry weather flow are another means to quantify I&I for sanitary sewer design. For example, Hong Kong uses peaking factors derived from catchment specific for flow monitoring which range from 1.00 to 1.50 to represent I&I. These factors are added to per capita flows values which have also been developed on a catchment specific basis (Yiu, 2005, p. 11).

³¹ Values are independent of peaking factors.

³² Peaking factors from 1.7 for 500,000 people to 7.8 for 30 people are applied to the per capita flows of either 210 L/cap-d for traditional design and 150 L/cap-d for "nu-sewer" design (Queensland Urban Utilities, 2010)

Table 9 Comparison of Residential Sewer Design Values

Municipality/Jurisdiction	Residential Sewer Design	Includes I&I Allowance	Municipality/Jurisdiction	Residential Sewer Design	Includes I&I Allowance
City of Calgary (Inflow and Infiltration Process Task Force, 2008)	380 L/cap·d	No	Queensland (Queensland Urban Utilities, 2010)	210 L/ cap·d	No
City of Chilliwack (Inflow and Infiltration Process Task Force, 2008)	410 L/cap·d	No	City of Saskatoon (Inflow and Infiltration Process Task Force, 2008)	290 L/cap·d	No
France (Marchés Publics de Travaux, 2003, p. 31)	200 to 250 L/cap·d	Yes	Canton du Valais (Canton du Valais, 2013, pp. 17-18)	250 L/cap·d	Yes
Hong Kong SAR (Yiu, 2005, p. 8)	190 to 350 L/cap·d	No	City of Victoria (Inflow and Infiltration Process Task Force, 2008)	225 L/cap·d	No
Greater Sudbury (Greater Sudbury, 2012, p. 8)	360 to 500 L/cap·d	No	Region of York (Inflow and Infiltration Process Task Force, 2008)	265 L/cap·d	No
Metro Vancouver (Engineering & Construction Department, 2005)	320 L/cap·d	No ³³	Auckland, NZ ³⁴ (Watercare Services Ltd, 2012)	ADWF 225 L/cap·d PWWF 1500 L/cap·d	No Yes

Swiss criteria for wastewater systems discharging into Lake Geneva use colour codes to flag estimated excessive I&I: “blue” for an equivalent per capita total flow less than 250 L/ cap·d, “green” for an equivalent per capita total flow between 250 and 450 L/ cap·d, and “red” for total flows above 250 L/ cap·d (Canton de Valais, 2013, p. 17). Under the “blue” criterion, 80 L/cap·d is allocated for I&I (Canton du Valais, 2013, p. 18). However, this system works as a response plan trigger and is not a design criterion: it illustrates how I&I is addressed differently by the Swiss.

Furthermore, to illustrate the Swiss approach, the Valais wastewater treatment plants reported high I&I after four dry years as 485 L/cap·d (Canton du Valais, 2013), exceeding the “red” threshold. At the same time, Valais also reports I&I increased in 2012 by 65% (Canton du Valais, 2013). The excess I&I triggered I&I remediation under Swiss regulations³⁵ which require sewer catchments to be rehabilitate to reduce I&I when the amount of excess flow is above 30% average dry weather flow (Kanton Zürich, 2004). Literature quantifying and linking I&I to return periods or precipitation was not found which suggests that the Swiss manage I&I independently of precipitation return periods.

³³ Revised from the original Rawn sanitary system design criterion of 95 gpcd or 430 L/cap·d which included allowance for I&I (Rawn, Hyde, & Oliver, 1953, p. 132)

³⁴ Default design parameters from WaterCare, a public water and sewer utility owned by Auckland Council. Sewer design procedures assume 3 persons per dwelling

³⁵ Art. 76 GSchG sowie Art. 5 Gewässerschutzverordnung, GSchV (Kanton Zürich, 2004)

Percentage of Rainfall I&I

An alternate design approach to account for I&I is used in Sydney, Australia and is based on rainfall. Sydney Water has set their I&I target in new sewers as corresponding 2% water ingress 30 years following construction of the sewer works (Sydney Water, 2012). Sydney Water's sewer capacity design is based on PDWF which includes an I&I allowance and the I&I design allowance is taken as 2% rainwater ingress over the serviced catchment as determined by 10 year simulation modelling, and is inferred to equal as an approximation of 2 times PDWF just for I&I. However, for new construction, the measured I&I rate is expected to be zero. Sewer condition inspection is identified on a 25 year cycle.

Appendix C: 2013 Estimated Financial Implications of Doubling the Inflow and Infiltration Allowance

Memorandum of February 18, 2013 (7061912)

From Taban Sowlati, Ph.D., P.Eng., Senior Project Engineer, Utility Planning Department, Metro Vancouver

To Brent Burton, M.A.Sc., P.Eng., Senior Engineer, Utility Planning Department, Metro Vancouver,



metrovancover

SERVICES AND SOLUTIONS FOR A LIVABLE REGION

MEMORANDUM

Greater Vancouver Regional District • Greater Vancouver Water District • Greater Vancouver Sewerage and Drainage District • Metro Vancouver Housing Corporation

Metro Vancouver
Utility Planning Department
Tel. 604-431-6159 Fax 604-435-6811

TO: Brent Burton

DATE: February 18, 2013

From: Taban Sowlati

File No.

cc: Yao-Hung Lan, Yvonne Wai

Re: ESTIMATED FINANCIAL IMPLICATIONS OF DOUBLING INFLOW AND INFILTRATION
ALLOWANCE FOR REGIONAL SEWERS

I. PURPOSE

In this memorandum, the theoretical impact on the scheduling of capital infrastructure of doubling the current regional I&I allowance is estimated. These estimates have been prepared based upon the results of several pilot studies completed throughout 2012. This information on the estimated implications for the scheduling of capital works is then used, in consultation with economists at Metro Vancouver, to estimate the financial implications.

II. BACKGROUND

Basic Service

The 2002 version of the Liquid Waste Management Plan specified a definition for Basic Service Flow (BSF) as estimated *peak dry* weather flows plus a wet weather allowance for inflow and infiltration of 11,200 l/ha/day. In the Integrated Liquid Waste and Resource Management Plan dated 2010, a similar definition is presented (Section 1.3.2) where a condition is noted to “maintain trunk sanitary sewer capacity for dry weather sewerage conveyance levels plus the Metro Vancouver target inflow and infiltration allowance; as necessary upgrade trunk sewer systems to maintain hydraulic grade lines and safe operating levels which have been established based on measured flow.”

Doubling I&I Allowance

In 2012, the REAC Liquid Waste Subcommittee (LWSC) requested that Metro Vancouver study the impact of doubling the I&I allowance in BSF to 22,400 L/ha/day. In support of this request, the UA Division prepared a variety of supporting analysis.

Three pilot studies in the Fraser Sewerage Area (FSA), one in the Lulu Island West Sewerage Area (LIWSA) and two in the North Shore Sewerage Area (NSSA) have been completed and presented to the REAC LWSC at various meetings throughout 2012. The pilot studies include:

- South Surrey Interceptor in South FSA;
- Coquitlam Interceptor in North FSA;
- Annacis Treatment Plant in FSA;

ESTIMATED FINANCIAL IMPLICATIONS OF DOUBLING INFLOW AND INFILTRATION ALLOWANCE FOR REGIONAL SEWERS
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- Gilbert Trunk Sewer in LIWSA; and
- Hollyburn Interceptor and North Vancouver Interceptor in NSSA.

Tables A1 to A6 in the Appendix summarize the estimated service life reduction for the pilot studies with increased I&I allowance. It is concluded that growth projects would be expedited on average, if the allowable I&I were doubled, by:

- 13 years in FSA;
- 24 years in LIWSA; and
- 18 years in NSSA.

III. Analysis and Cost Estimates in FSA

In order to calculate the financial implications of doubling I&I in Metro Vancouver sewers, the capital funds for all infrastructure growth projects have been considered based on the 2013 Utility Long Range Plan (LRP) version 8.6. Table 1 summarizes the estimated growth capital funds in FSA from 2013 to 2051. The followings are the key assumptions incorporated in the analysis:

Projects identified on the 2013 LRP (2013-2022):

- Fund all growth projects based on the 2013 Utility LRP version 8.6.
- Treatment plant project costs are excluded.

Projects identified beyond 2022:

- Project upgrades are estimated based upon long range Basic Service projections using existing criteria (i.e. maintain capacity for estimated PDWF plus allowable I&I of 11,200 L/ha/day)
- Project upgrades beyond 2022 are identified across South FSA. The South FSA pilot area is from East Langley to Annacis Island Treatment Plant and also from Katzie Pump Station to the North Surrey Interceptor - River Crossing Section.
- Project funding beyond 2022 in South FSA estimated based on the unit cost of \$25/linear foot/inch diameter and additional design and contingency funds provided by E&C in 2012.
- Then, the estimated funds for the entire FSA beyond 2022 used the projected funds in South FSA as a basis for the calculation.
- Population and landuse projections are provided by Regional Development Division.

Table 1. Estimated growth capital funds in FSA from 2013 to 2051 based on the undiscounted cash cost:

	Total Capital costs (\$) for Growth Category projects	Average required funds (\$) per year*
2013 to 2022	447,371,000	Refer to 2013 LRP version 8.6
2023 to 2031	82,416,416	9,157,380
2032 to 2041	174,030,631	17,403,063
2042 to 2051	13,017,079	1,301,708

ESTIMATED FINANCIAL IMPLICATIONS OF DOUBLING INFLOW AND INFILTRATION ALLOWANCE FOR REGIONAL SEWERS
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*it is assumed that funds are uniformly distributed in a decade

Based on the information presented in Table A1 and A2 of the appendix, capital works in FSA would need to be advanced by an average of 13 years in case of twice I&I scenario. Table 2 summarizes the required undiscounted capital funds for each decade from 2013 to 2051. As shown, the cost for capital projects would increase by 34% from 505.6 Million to 676.6 Million by year 2022.

Table 2: Comparison of required undiscounted growth funds in FSA with and without I&I increase between 2013 and 2051

	Total Growth funds without I&I increase	Total Growth funds with I&I increase	Increase (%)
2013 to 2022	447,371,000	599,399,668	34%
2023 to 2031	82,416,416	108,323,502	31%
2032 to 2041	174,030,631	9,111,955	-95%*
2042 to 2051	13,017,079	N/A	N/A

* may not be representative due to lack of information on Capital works beyond 2051.

Table 3 summarizes the effect of doubling I&I between 2013 and 2041 in more details by comparing required funds per year based on both the undiscounted and discounted projected capital funds. Please note that the real discount rate of 4%, provided by the economist at Metro Vancouver, is applied to the estimated funds. The undiscounted funds without I&I increase (second column) is based on the total growth costs indicated in the 2013 LRP version 8.6. As mentioned before, the allocated funds for the wastewater treatment projects are excluded.

If all the projects are to be advanced theoretically by 13 years, the majority of capital items need to be completed in 2013. Therefore, the required fund would be \$475.5 Million.

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Table 3: Comparison of required undiscounted and discounted growth funds in FSA with and without I&I increase between 2013 and 2041

Year	Undiscounted value at \$25/inch/foot length		Net Present values (base year 2013); real discount rate: 4%	
	Growth funds (\$) without I&I increase	Growth funds (\$) with I&I increase	Growth funds (\$) without I&I increase	Growth funds (\$) with I&I increase
2013	22,890,000	475,754,835	22,890,000	475,754,835
2014	32,580,000	9,157,380	31,307,692	8,805,173
2015	44,100,000	9,157,380	40,772,929	8,466,512
2016	80,721,000	9,157,380	71,760,675	8,140,877
2017	58,200,000	9,157,380	49,749,604	7,827,766
2018	28,775,000	17,403,063	22,007,098	14,304,049
2019	48,100,000	17,403,063	36,433,500	13,753,894
2020	84,470,000	17,403,063	64,190,258	13,224,898
2021	40,080,000	17,403,063	29,286,063	12,716,248
2022	11,475,000	17,403,063	8,062,183	12,227,161
2023	9,157,380	17,403,063	6,186,398	11,756,886
2024	9,157,380	17,403,063	5,948,459	11,304,698
2025	9,157,380	17,403,063	5,719,672	10,869,902
2026	9,157,380	17,403,063	5,499,685	10,451,829
2027	9,157,380	17,403,063	5,288,159	10,049,835
2028	9,157,380	17,403,063	5,084,768	9,663,303
2029	9,157,380	1,301,708	4,889,200	694,992
2030	9,157,380	1,301,708	4,701,154	668,262
2031	9,157,380	1,301,708	4,520,340	642,560
2032	17,403,063	911,195	8,260,232	432,492
2033	17,403,063	911,195	7,942,531	415,858
2034	17,403,063	911,195	7,637,049	399,863
2035	17,403,063	911,195	7,343,316	384,484
2036	17,403,063	911,195	7,060,881	369,696
2037	17,403,063	911,195	6,789,309	355,477
2038	17,403,063	911,195	6,528,181	341,805
2039	17,403,063	911,195	6,277,097	328,658
2040	17,403,063	911,195	6,035,671	316,018
2041	17,403,063	911,195	5,803,529	303,863
Total			493,975,632	644,971,891

ESTIMATED FINANCIAL IMPLICATIONS OF DOUBLING INFLOW AND INFILTRATION ALLOWANCE FOR REGIONAL SEWERS
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Table 4 summarizes the Net Present Value (NPV) without I&I increase and with I&I increase between 2013 and 2022 and also between 2013 and 2041 based on the 4% real discount rate. The difference in calculated NPV until 2022 is almost \$200 Million and until 2041 is about \$150 Million.

Table 4: Comparison of NPV with and without I&I increase

	NPV without I&I increase	NPV with I&I increase	Difference	increase %
Total NPV until 2022	\$ 376,460,002	\$ 575,221,413	\$ 198,761,411	53%
Total NPV until 2041	\$ 493,975,632	\$ 644,971,891	\$ 150,996,259	31%

IV. Analysis and Cost Estimates in LIWSA

Based on the information presented in Table A3 of the appendix, capital works in LIWSA would need to be advanced by an average of 24 years in case of twice I&I scenario. The Capital funds for the Gilbert/Brighthouse Trunk Sewer twinning Phase 1 to 4, part of the Infrastructure Maintenance category, are indicated in the 2013 Utility LRP version 8.6. The corresponding allocated funds are about \$97 Million for the period between 2013 and 2020. In case of twice I&I allowance, the projected fund would be immediately required in 2013. Therefore, the 2013 capital cost would be increased drastically from \$8.14 Million to \$97.0 Million as shown in Table 4.

Table 4: 2013 Cost allocation for Gilbert/Brighthouse with and without I&I increase

Gilbert/Brighthouse Trunk Sewer Twinning Phase 1 to Phase 4 in year 2013	
Total cost without I&I increase	\$8.14 Million
Total cost with I&I increase	\$97.0 Million
Cost increase	\$88.86 Million

V. Conclusions

Based on information contained in this memorandum, the theoretical financial implications due to doubling the allowable I&I in the standard BSF are ESTIMATED as follows:

- Growth projects would be expedited on average by 13 years in FSA.
- The undiscounted Capital costs related to FSA growth projects scheduled between 2013 and 2022 would increase by 34% from \$447.3 Million to \$599.4 Million.
- The NPV of the estimated Capital costs related to FSA growth projects between 2013 and 2022 would increase drastically from \$376.4 Million to \$575.2 Million, an increase of 53%, based on the real discount rate of 4% and if all the projects are advanced by 13 years.
- The NPV of the estimated Capital costs related to FSA growth projects between 2013 and 2041 would increase drastically from \$494 Million to \$645 Million, an increase of 31%, based on the real discount rate of 4% and if all the projects are advanced by 13 years.

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- Capital projects in LIWSA would be required immediately.
- Capital costs related to LIWSA Maintenance projects scheduled between 2013 and 2020 would increase from \$8.14 Million to \$97 Million.

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APPENDIX:

Table 1A: Estimated Effect of Doubling I&I on Scheduling of Basic Service Upgrading for the South Surrey Interceptor; presented at REAC Liquid Waste meeting on Aug. 16, 2012

South Surrey Interceptor Sewer Section		Section Upgrade		Reduction in Service Life
		With I&I =11,200 L/ha/day (no increase)	With I&I =22,400 L/ha/day (increase)	
1	River Crossings	2038	2012	26 years
2	Delta Section	2033	2018	15 years
3	Peck Road Section	2026	2024	2 years
4	Scott Road and Highway 91 sections	Upgrade requirement is beyond 2051 if downstream sections are upsized as stated above	Upgrade requirement is beyond 2051 if downstream sections are upsized as stated above.	—
5	Panorama and Colebrook sections	Upgrade requirement is beyond 2051 if downstream sections are upsized as stated above.	Upgrade requirement is beyond 2051 if downstream sections are upsized as stated above.	—
6	Johnston Section	2019	2012	6-7 years

ESTIMATED FINANCIAL IMPLICATIONS OF DOUBLING INFLOW AND INFILTRATION ALLOWANCE FOR REGIONAL SEWERS
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Table 2A: Estimated Effect of Doubling I&I on Scheduling of Basic Service Upgrading for the North Vancouver Interceptor; presented at REAC Liquid Waste meeting on Aug. 16, 2012

North Vancouver Interceptor Sewer Section		Section Upgrade		Estimated Reduction in Service Life
		With I&I =11,200 L/ha/day (no increase)	With I&I =22,400 L/ha/day (increase)	
1	Deep Cove Section Extension	>2051	12051	-----
2	Deep Cove Section	>2051	>2051	-----
3	Seymour Section	>2051	>2051	-----
4a	City Section - Upstream	>2051	2031	20 years
4b	City Section - Downstream	>2051	>2051*	-----*

*Requires replacement of a section of Wainwright Road Interceptor LWW Midway Creek

Table 3A: Estimated Effect of Doubling I&I on Scheduling of Basic Service Upgrading for the Gilbert Trunk; presented at REAC Liquid Waste meeting on Oct. 18, 2012

Bridgeport PS to Lulu Island Treatment Plant		Section Upgrade		Estimated Reduction in Service Life
		With I&I =11,200 L/ha/day (no increase)	With I&I =22,400 L/ha/day (increase)	
1	Brighthouse Trunk Sewer-South	2028	2012	16 years
2	Gilbert Trunk	2043	2012	31 years

ESTIMATED FINANCIAL IMPLICATIONS OF DOUBLING INFLOW AND INFILTRATION ALLOWANCE FOR REGIONAL SEWERS
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Table 4A: Estimated Effect of Doubling I&I on Scheduling of Basic Service Upgrading for the Hollyburn Interceptor; presented at REAC Liquid Waste meeting on Oct. 18, 2012

Hollyburn Interceptor Sewer Section		Section Upgrade		Estimated Reduction in Service Life
		With I&I = 11,200 L/ha/day (no increase)	With I&I = 22,400 L/ha/day (increase)	
1	28 th St. to Kew Rd. Section	>2031	2012	19 years
2	Looped Section	>2026*	2012	15 years
3	Hollyburn Relief Sewer	>2026*	2012	15 years
4	Hollyburn Interceptor (limited sections)	>2026*	2012	15 years

*Strategic flow monitoring completed / planned to better estimate the required timing of capacity upgrading for the base case.

Table 5A: Estimated Effect of Doubling I&I on Scheduling of Basic Service Upgrading for the Annacis Island Treatment Plant; presented at REAC Liquid Waste meeting on Oct. 18, 2012

- Total Existing Basic Service Flow: Approximately 12.5 m³/s
- Total Existing Basic Service with Doubling of I&I Allowance: Approximately 18.9 m³/s
- Result:
 - Exceeds PWWF capacity of Stage V (15.8 m³/s)
 - Immediate upgrade to Stage VI (capacity = 18.9 m³/s)
 - Upgrade shortly thereafter to Stage VII (capacity = 22.14 m³/s)

Table 6A: Estimated Effect of Doubling I&I on Scheduling of Basic Service Upgrading for the Coquitlam Interceptor; presented at REAC Liquid Waste meeting on Nov. 15, 2012

Coquitlam Interceptor		Section Upgrade		Estimated Reduction in Service Life
		With I&I = 11,200 L/ha/day (no increase)	With I&I = 22,400 L/ha/day (increase)	
1	Port Moody Section	beyond 2051	2048	~ 5 years
2	Dewdney Section	beyond 2051	2031	20 years

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3	Lougheed Section	beyond 2051	2033	18 years
4	Cape Horn Section	beyond 2051	beyond 2051	-

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Appendix D: City of Coquitlam: Inflow and Infiltration Management Strategy

Memorandum of September 19, 2013 (9583067)

From Dana Soong, P.Eng., Manager of Utilities Programs, City of Coquitlam

To Regional Engineering Advisory Committee – Liquid Waste Subcommittee

Coquitlam

Memo

September 19, 2013
Our File: 11-5340-01/000/2013-1
Doc #: 1543193.v1

To: REAC Liquid Waste Subcommittee
From: Dana Soong, P.Eng., Manager Utility Programs

Subject: Coquitlam's Inflow & Infiltration Management Strategy

As part of the region's 2010 Integrated Liquid Waste and Resource Management Plan, the City committed to developing and implementing an inflow & infiltration (I&I) management plan to ensure wet weather flows are within acceptable allowances. This briefing memo provides a high level synopsis of the City's I&I management strategy.

For I&I management purposes, the City can be generalized into three sectors with characteristics shown below (Attachment A).

Region	Length of Sewers (kms)	Area (ha)	Predominate Age	Predominate Pipe Material	Estimated I&I (L/ha/d)
Northwest Sector	154	1,550	1980/90's	PVC	16,400 to 17,500
Southwest Sector	279	3,390	1960/70's	Conc, AC, VC	35,800 to 44,000
Northeast Sector	37	1,070	2000's	PVC	not applicable

The SW sector has the oldest sewers and highest estimated I&I (Attachment B). This area also went through a ditch elimination program in the 1990's, which may explain why several cross connected properties have been identified.

Estimates of I&I rates were made using Metro Vancouver's graphical method. However, the 5 year I&I values were linearly extrapolated from small rain events and are likely exaggerated. The initial focus of Coquitlam's management plan is to get accurate estimates of I&I.

File #: 11-5340-01/000/2013-1 Doc #: 1543193.v1

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September 19, 2013

Discussion:

Our strategy for managing I&I in the sewer system consists of quantifying the magnitude of the problem, identifying sources of I&I and reducing I&I at the source.

Quantifying Magnitude

- Install permanent flow monitors in sectors of concern (planning to install 2 in NW sector and 2 in SW sector)

Identifying Sources

- CCTV of sewer main – Annually inspecting about 3% of system (SW sector completely inspected in 1990's, NW sector completed in 1990/2000's)
- CCTV of service laterals – Inspected as part of the paving rehabilitation program, about 1.2% per year (PVC pipes are not currently inspected). Program includes installation of IC's at property line and video inspection of both the public and private lateral
- Smoke testing catchments with high I&I

Reducing I&I

- Repair of any defects identified in sewer mains and public service laterals, including repairs of joints with visible infiltration.
- Notify property owners of defects on private laterals
- Rectify cross connections (storm to sanitary) – property owners are required to correct problem or disconnect roof drainage from sewer system.

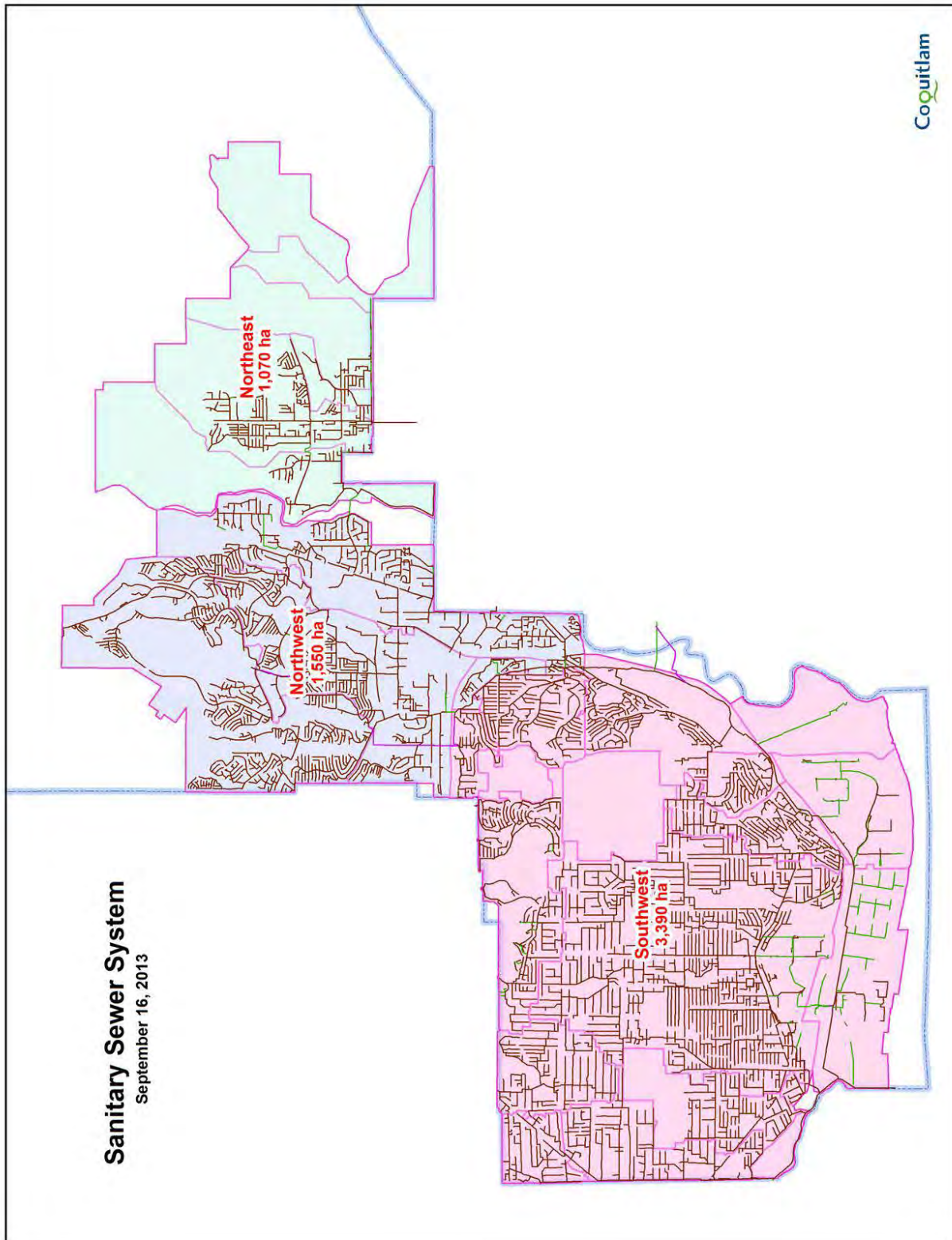
Management of I&I in Coquitlam is considered an ongoing, long range program where reductions will be achieved through repairs, asset replacement and building redevelopment.

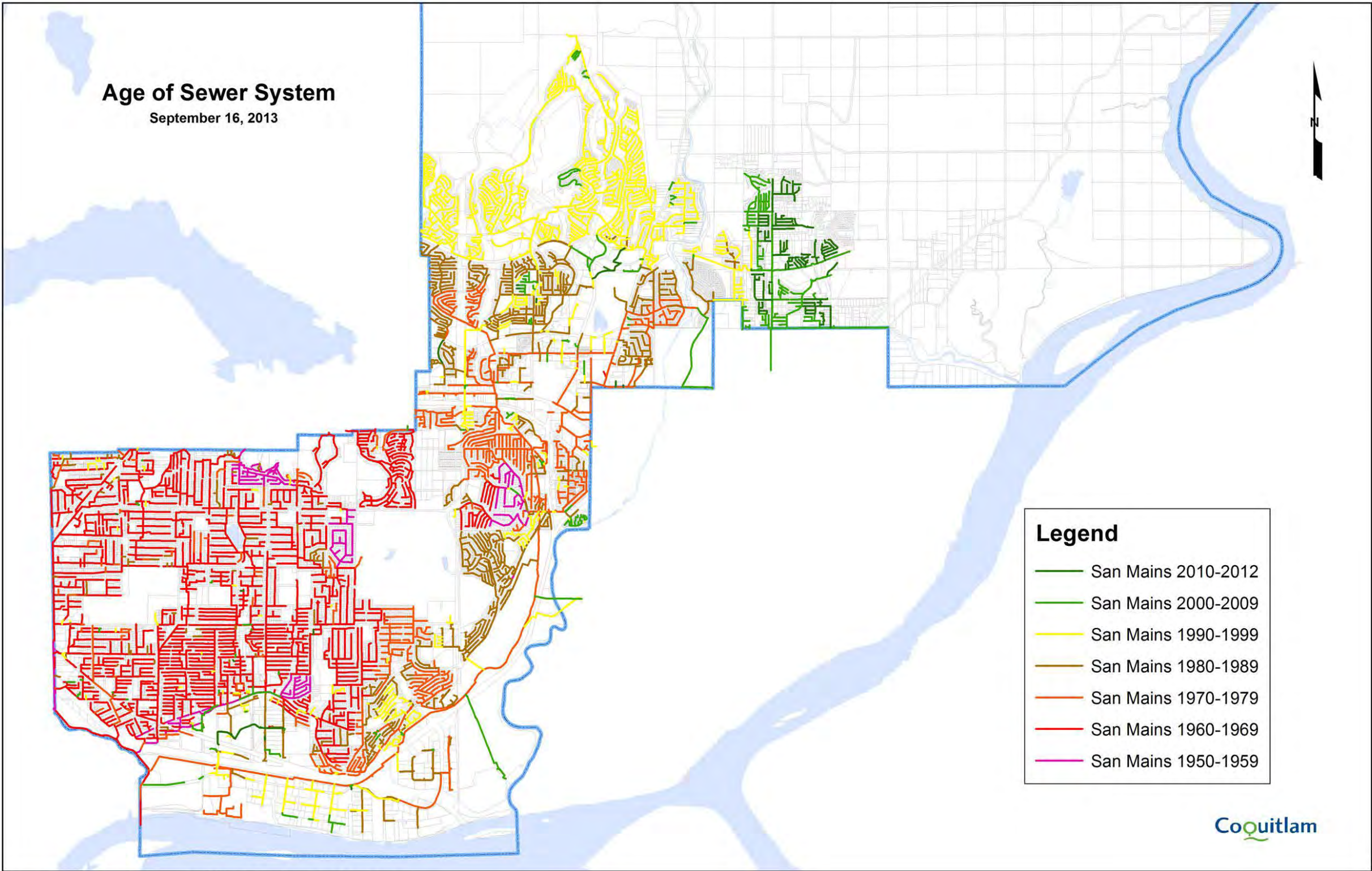


Dana Soong, P.Eng.

Attachments:

- A. Map of I&I Sectors in Coquitlam
- B. Map showing Age of Sewers





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Appendix E: District of North Vancouver: Inflow & Infiltration Reduction Program 2014 Status Report

E-mail of April 29, 2014 (9582997)

From Shaun Carroll, P.Eng., Section Manager, Utilities Technical Services, District of North Vancouver

To Regional Engineering Advisory Committee – Liquid Waste Subcommittee



Inflow & Infiltration Reduction Program 2014 Status Report

District of North Vancouver

April 2014

**Shaun Carroll
Section Manager,
Utilities Technical Services**

Document: 2319347

DNV Inflow & Infiltration Reduction Program
2014 Status Report

April, 2014
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Document: 2319347

**DNV Inflow & Infiltration Reduction Program
2014 Status Report**

**April, 2014
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On December 14, 2011 representatives from the Ministry of Environment (MOE) and Metro Vancouver (MV) met with the District of North Vancouver (DNV) to discuss the formalization of DNV's Inflow and Infiltration (I&I) reduction program, focused initially in the DNV's Lynn Valley area.

This 2014 report tracks both historic work and describes future planned work within the District of North Vancouver.

1 Background

The May 2012 ILWRMP action item 1.2.4 mandates that chronic sanitary sewer overflows (SSOs) be reduced to a frequency of less than once every 5 years for a 24 hour storm event. The Lynn Branch, located within the DNV, has recorded overflows related to I&I from the DNV's Lynn Valley sewershed. These overflows have occurred from rainfall events of less than the 5 year, 24 hour event.

The DNV is working on a Lynn Valley sewershed I&I reduction program which we expect will reduce future LBISB SSO's to mandated allowable frequencies. The reasons for DNV's focused strategy include the following:

1. Eliminate the need for a MV's \$15M concrete sewage retention tank at the Lynn Branch inverted siphon, required by the MOE to eliminate the 5 year 24 hour event,
2. Fulfill our ILWRMP responsibility by taking measures to reduce I&I and maintain our infrastructure,
3. Free up capacity in our sanitary sewer system to accommodate OCP forecasted population growth,
4. Identify local cost to benefit relationship for I&I reduction programs and effective removal rates,
5. Reduce wet weather sewage flow in anticipation of a potential revision to Metro's sewage rate structure, and
6. Deepen our understanding of DNV's long term sanitary sewer loading that will assist MV in designing the planned new North Shore Waste Water Treatment plant.

2 Budget Commitment

The DNV Lynn Valley I&I reduction program had a 2012 budget of \$650K, a 2013 budget of \$1.5M, and similar levels for 2014 and the following 3 years resulting in about \$8M over 5 years.

3 I&I Components Progress

Progress continues in a number of areas of our I&I program including Sewer Bylaw amendments, education, internal procedures, data analysis, and field work.

3.1 Sewer Bylaw Amendment and Bylaw Work

The DNV's I&I program is working closely with the residents on ongoing private property work and we have gained council support for key program elements related to this work. In 2012 we received Council agreement to pilot a private side sewer lateral remediation program. The

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expectation is that for the fall of 2014, we will be forwarding a permanent bylaw amendment that will allow the DNV to work on private side I&I issues.

Also in 2012, Council approved an updated Sewer Bylaw to require the homeowner/developer to fund the public side sewer lateral replacement/rehabilitation as part of the building permit process where the building permit value is over \$150,000. As a result of this, in 2013, a total of 37 public laterals and lateral interfaces were rehabilitated using CIPP liners, injection joint grouting and lateral interface injection grouting.

DNV has participated in a Metro initiative to examine the potential for Municipalities to require lateral I&I removal work at the time of property sale. The report "An Approach Toward Private Sewer Lateral Certification in Real Estate Transactions (Pinna Sustainability Inc, Civis Law & Stantec Consulting, 2013) resulted. As a result of this work Metro is pursuing an agreement from the real estate industry to include sewer laterals on a volunteer homeowner checklist at time of sale.

3.2 Education

In 2012 Utilities completed three educational workshops to inform council on the impacts of I&I on the DNV and Metro system and the benefits of performing and funding I&I reduction works, including works on private property. Utilities also trained internal staff on procedures and practices for the smoke and Dye testing of laterals.

In 2013 multiple internal training initiatives were completed. A series of three presentations was made to all DNV Utilities staff with the intent of increasing awareness and consequence of I&I, our motives for pursuing I&I reduction and updates on the programs that are currently in place. A select group of maintenance and development field staff received training on lateral pipe bursting, lateral CIPP patch repairs and lateral interface connection methods and products.

Council will be given an I&I status workshop in the fall of 2014 to allow the Sewer Bylaw amendment to be fully understood prior to requesting approval, which is scheduled for Dec 2014.

3.3 Internal Procedures

In 2012 we began reviewing our construction practices for sanitary and storm sewer work processes to ensure current construction, operation and maintenance practices support I&I abatement and reduction. Initiatives started in 2012 also include:

- Purchase of products and trained field staff on injection grouting techniques for stopping infiltration from manholes.
- Purchased products and trained staff on cured in place pipe (CIPP) point repairs for rehabilitating and reducing infiltration from sewer laterals.

In 2013 new products and procedures were put in place to reduce infiltration from laterals.

- New lateral interface connection methods were introduced and put into practice. The new methods and products include the exclusive use of manufactured fittings with gasket seals to reduce the likelihood of lateral interface infiltration and structural failure of new connections and joints.
- Lateral pipe bursting equipment was purchased, staff were trained on bursting procedures and we began replacing laterals with this technique.
- Utilities worked with DNV Development Services department to support the enforcement of MMCD inspection and testing requirements for new construction (mains and laterals).

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The inspection and testing requirements help to ensure I&I in new construction is within the allowable design limit

In 2013 we also created smoke testing field procedures and a test result documentation tool. A maintenance crew was established, trained and dedicated to locating potential cross connections in the Lynn Valley sewershed using smoke testing.

Our 2014 plan will focus on revising our DNV Design Guidelines for sewer lateral connection and developing Dye Testing methods and documentation protocols.

The DNV is an active participant in the annual National Benchmarking Initiative I&I Task Force and has performed two annual wastewater utility benchmarking cycles.

3.4 Data Analysis

A key deliverable of the I&I reduction strategy is the determination of effective I&I reduction rates so we can lay the foundation for efficient and cost effective I&I removal in the balance of the DNV and other North Shore Municipalities. To do this, the DNV has progressed toward key milestones with I&I analysis in 2013. We plan to quantify and monitor I&I consistent with Metro's I&I Management Plan Template.

In 2013, we completed our sanitary sewer model to assist in prioritizing our I&I reduction approach. Further updates to the model occur as required and several updates each year would be expected.

We are initially focusing on areas that contribute to sewage lift stations because analysis techniques finalized in 2012 allow us to use sewage lift stations as flow monitors for I&I analysis. This is significant as our historic SCADA data now allows for substantial historic flow data to contribute to the analysis of effectiveness of our I&I removal strategies. Once we are confident that an I&I removal technique is effective and cost justified or worth further evaluation, we are plan to apply the techniques to increasing catchment sizes for strategy verification purposes.

3.4.1 Crestlynn Analysis

Work continues in our small (26 SFR homes) Crestlynn catchment and costs associated with the Crestlynn I&I removal continue to change due to ongoing work. We have measured a significant reduction in inflow from this catchment. A graph that shows flow from the Crestlynn catchment, after 6 inflow sources (private side) have been redirected to the storm system, has been attached and indicates significant peak inflow removal.

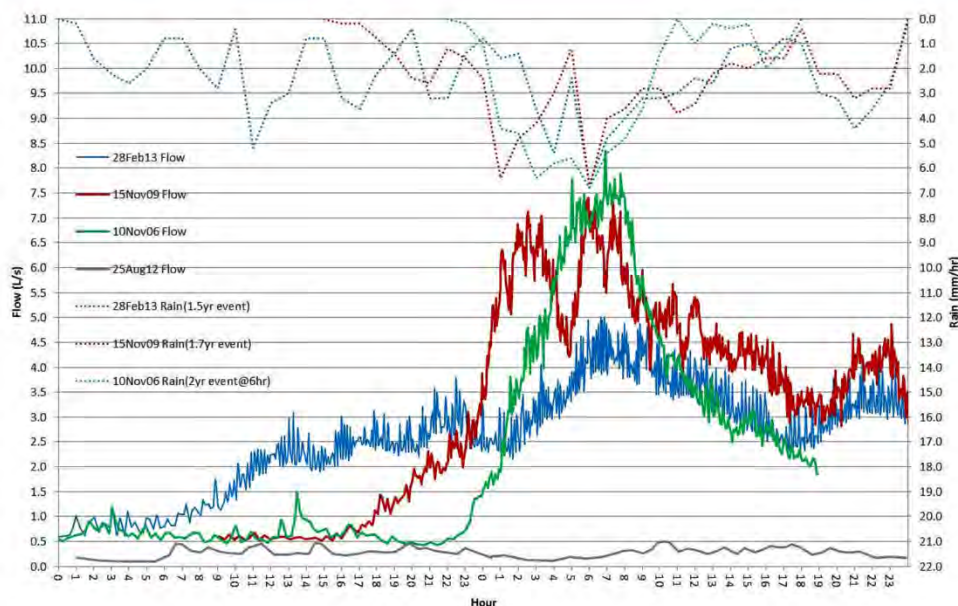
In the following graph the dotted lines at the top are rainfall and reference the scale on the right hand side (mm/h). The blue dotted line is a two year event for rain for Feb 2013 (after Crestlynn cross connection removal), while red and green are near two year events prior to cross connection removals. The rain events have similar peaks, which is relevant to inflow rates.

At the bottom of the graph there are three lines, a grey dry weather flow, and red and green pre cross-connection removal flows, and blue post cross-connection removal flow. What is apparent is the removal of the peak of the curve, or the inflow. Based on similar events, it appears we have removed between about 2.5-3.5 L/s of inflow (from 6 cross connections) equalling about a 0.5 L/s per property flow for a 2 year event.

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3.4.2 General Analysis

Additional analysis has highlighted some key issues related to the percentage split of the infiltration and inflow components from any rain event. Once known for a particular rainfall event, the inflow component of sewage in a rainfall event can be estimated, with a relatively high confidence, for varying intensities and durations based on the fact that rainfall entering through a direct connection to the sanitary system has little physical reason to vary. The contributing flow will therefore be closely related to the intensity of the rain event. However, infiltration will not create a linear response to rainfall (like inflow) due to the mechanism of entry into the sanitary sewer system which is directly related to the extremely significant antecedent rainfall and associated soil saturation conditions. As an example a flash 2 year return 15 minute rainfall in the middle of summer will have a nominal infiltration impact. That same event immediately following a large winter storm may cause significant infiltration to be contributed to the sanitary system.

Due to this rationale, it is extremely difficult for us to predict what flow rate comparison (and more importantly the flow percentage reduction) will be under a 5 year event, if a 2 year event is the only reference point. For this reason, we are only comfortable comparing flows from similar rainfall events (in similar seasons) when evaluating success of certain remediation techniques, which can substantially slow the evaluation process. Fortunately, we have a significant history with our lift stations acting as flow meters.

We have completed additional infiltration reduction measures, in the form of basin-wide connection interface remediation, within the Crestlynn catchment area but have yet to

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experience a comparable event to gauge the success of that effort. Definitive success criteria will be based on like rainfall events.

3.5 Cost Analysis

Based on our Crestlynn inflow removal experience and using our recorded costs to remove the cross connections, we created a preliminary \$/L peak flow removal estimate. A significant cost of installing a storm sewer is included in the high end of expected L/s flow removal. Average costs per property, and per liter inflow removed, are shown below.

\$/property Inflow Removal (Including storm sewer install costs)	\$21,000	\$/Litre Inflow Removal (Including storm sewer install costs)	\$39,000
\$/property Inflow Removal (Connection work only)	\$6,500	\$/litre Inflow Removal (Connection work only)	\$13,000

It should be stressed that these numbers are based on a small data set and inflow removal only. An integral component of the I&I reduction program is to expand the data set with the hope of validating these values.

In 2013 our focus in Lynn Valley was on smoke testing and IC installation in preparation for dye testing. We completed smoke testing of approximately 5000 SFR laterals for a program cost of \$47k (approximately \$9.50/property). The results of the 2013 smoke testing program indicate that we have 68 SFR properties that are potentially cross connected. These locations are our first priority for the 2014 Cross Connection Removal Program where we will use a combination of Dye Testing and Lateral CCTV Inspection to confirm cross connection locations. To facilitate the dye testing we have installed 49 inspection chambers to date at a total cost of \$98k (approximately \$2000/property). In 2014 we will complete 19 more IC installations and perform bulk dye testing in the Lynn Valley sewershed. We are planning to remediate cross connections in 2014/2015 and will move forward with other I&I reduction initiatives while hoping to see rainfall events that will help us quantify the benefit from each remediation approach.

The 2013 smoke testing results suggest that 2% of the laterals are cross connected. The expected removal rate per cross connected property for a 2 year event is 0.5 L/SFR/cc. There are about 5500 sanitary laterals in the Lynn Valley sewershed therefore it is we believe it is unlikely that we will discover enough cross connections to able to remove our target 150 L/s peak inflow removal for a 5 year 24 hour event using only smoke testing to locate cross connections.

We have preliminary data that suggests that dye testing will identify cross connections that were not identified by smoke testing. If dye testing proves effective at locating cross connections then costs for private side/public side works could range from \$2.0M (no pumping required) to \$5.9M (pumping half storm connections). If this is not the case then infiltration remediation will become the key approach to remove the balance of flow reduction not addressed through the cross connection removal and inflow reduction program.

It is important to note that, in the Crestlynn experience, an unexpected cost was the construction of a new storm main was constructed to service three adjacent SFR properties. The capital cost of the new storm main was estimated to be approximately equal to the cost capital cost to install three private pump stations. This additional work component added significant costs to this program. As the program expands we will be able to more accurately

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identify if this is a normal program cost or if the Crestlynn experience was an anomaly in the DNV.

We are collecting valuable costs for work related to system inspections required to identify I&I sources. Costs, based on 2013 quantities and costs information, are as follows:

Inspection Components	Units	# Completed	Cost (\$/unit)
Inspection Chamber install	Per connection	49	\$ 2000
Smoke Testing	Per connection	5000	\$ 9.50
Video Inspection (main)*	Per meter	17628	\$ 6.60
Manhole Inspection	Per MH	1066	\$20
Main Launch Lateral Insp.	Per connection	26	\$ 410
IC Launch Lateral Insp.	Per connection	37	\$ 95.00

*Includes all annual program costs including flushing, cleaning, debris removal and traffic control.

In 2014, we will be continuing work in the Lynn Valley catchment to allow us to complete infiltration and inflow removal initiatives and continue efforts to identify the related cost/benefit analysis. The cost/benefit analysis will then be used to focus I&I reduction efforts across the DNV.

3.6 2012-2013 Field Work

The I&I Lynn Valley field strategy initially focused on reducing peak flow through a number of program components noted below.

3.6.1 Sanitary Sewers

3.6.1.1 Plugging Manhole Lid Holes

A relatively simple program that involved plugging all holes in 1760 manhole lids in the Lynn Valley sewershed. This survey also collected data on the MH's inflow potential recording pavement defects that would increase the MH's likelihood of allowing infiltration.

3.6.1.2 Manhole Inspection and rehabilitation

In 2013 MH inspection data from 2011 and 2012 inspection contracts was amassed and analyzed for infiltration rehabilitation potential. In total 1066 MH inspections were analyzed resulting in 213 (20%) identified for infiltration abatement through injection grouting. The injection grouting rehabilitation contract will be completed in 2014.

3.6.1.3 Sanitary Sewer Rehabilitation

In 2013 the CCTV inspection results from the 2011 and 2012 CCCTV inspection, which focused in the Lynn Valley area, were compiled and analyzed to identify points of infiltration for rehabilitation. Four rehabilitation contracts were tendered as a result of the analysis.

1. A contract for pressure tests and injection grouting of joints in 3,443 m of main, 379 lateral interface connections and 13 point defects.
2. A contract for CIPP liners including 640 m in nine full length MH to MH liners and a combined 73m at 32 point liner locations.
3. Two contracts for external point repairs at 10 individual locations.

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Work on all four contracts will be completed in 2014 and pricing will be verified at that point.

3.6.1.4 Sanitary Sewer CCTV Inspection

By 2013 80 % of the Lynn Valley's sanitary sewer mains were inspected. Inspection of the remaining 20% was unsuccessful due to encroachment of public property either on MH's or inhibiting access to MH's. Efforts will be made to complete the inspection in 2014.

3.6.1.5 Lateral Interface Inspection and Rehabilitation

Lateral interface inspections occur during our annual contracted video inspection programs. Rehabilitation of lateral interface defects will occur as a component of the annual rehabilitation work.

3.6.2 Sanitary laterals

3.6.2.1 Lateral IC Inventory

Approximately 2000 of 4000 properties in Lynn Valley have been inspected for the existence of a sanitary IC chamber. This component was completed in 2012. We discovered that 10% of properties have a sanitary IC chamber installed.

3.6.2.2 Lateral IC Cover Inspection and Replacement

In 2012 staff completed a GIS mapping inventory of IC chambers and replaced missing cleanout covers. This data indicates that only 10% of the existing lateral inventory has. Our current development and sewer connection bylaws ensure that IC's are installed of new developments, renovations >150k, where new connections or temporary caps are required.

3.6.2.3 Installation of Sanitary Lateral ICs

In 2012 staff installed sanitary and storm IC's at all 26 properties in the Crestlynn catchment. In 2013 the staff and contractors installed 49 sanitary ICs in the Lynn Valley catchment. We estimate the need to install an additional 100 ICs to address cross connections in Lynn Valley.

3.6.2.4 Lateral Inspection Equipment Pilot Program

A main line launched lateral inspection was performed of each sanitary lateral in the Crestlynn catchment. The results directed us to invest in lateral launch camera technology to help identify I&I on the lateral which otherwise require a lateral inspection chamber. We requested funding and had funding approved for a lateral launch camera in 2013 for 2014 purchase.

3.6.2.5 Lateral Inspection Program

Trained internal staff are assessing for signs of I&I while completing operational CCTV inspections for lateral pipe inspection and rehabilitation. A second CCTV camera and MH access camera were purchased to support and increase efforts for 2012/2013.

3.6.2.6 Smoke and Dye Testing Program

In 2012 and 2013 the staff smoke tested all SFR laterals in Lynn Valley. In 2012 Crestlynn was dye tested. In 2012 six cross connections on the private side were removed in the Crestlynn sewershed. In 2012 and 2013 smoke testing identified 35 potential cross connections in the Birchlynn sewershed (sub-catchment in Lynn Valley) and another 37 in the Ross and Allen sewershed (sub-catchment in Lynn Valley). Efforts in 2014 will focus on cross connection confirmation and removal in the Birchlynn sewershed.

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One of our needs from the smoke and dye testing program is to help us identify how effective each method is at detecting and locating cross connections. Preliminary results have indicated that smoke testing can incorrectly identify cross connections and can miss identifying cross connections. We have also noted that dye test effectiveness results are very sensitive to weather conditions. For example, in Crestlynn we recently (2014) repeated dye testing on all 26 properties to check if our cross connection remediation work was still in place. The staff discovered 2 “new” cross connections that were not identified in the 2012 smoke program or the 2013 dye program. Our I&I reduction program will continue to explore these findings to identify the test method limitation and to help optimize the effectiveness of the I&I reduction program.

3.7 Planned 2014 Work

The 2014 plan will build and expand on the existing programs.

3.7.1.1 Bylaw Enhancement

The DNV Design Guidelines and Sewer Use Bylaw for lateral connection method and material is being updated to include connection method and hardware that is more I&I resistant than the current method and materials.

3.7.1.2 Plugging Manhole lids

Ongoing replacement, where necessary, of manhole lid hole plugs will continue.

3.7.1.3 Manhole Inspection and Rehabilitation

Out of the approximately 1,100 manholes inspected in 2011/2012, about 213 were rated as having infiltration scores of 3, 4, or 5 (where 5 is the worst). The manholes are located primarily in Lynn Valley. MH rehabilitation through injection grouting will take place in 2014.

3.7.1.4 Sanitary Sewer Rehabilitation

In the Lynn Valley area a contract valued at \$350k to perform rehabilitation through injection grouting of sanitary 3443 m of sewer main joints/defects and 379 lateral interface connections and will be completed in 2014.

In the Lynn Valley area a contract valued at \$340k to complete 640 m of full length and 73 m of point CIPP liners and will be completed in 2014.

In the Lynn Valley area two contracts valued at \$84k to complete external point repairs and will be completed in 2014.

The 2014/2015 sewer main rehabilitation program will include warranting pipes from the 18,000 m of main CCTV inspected in 2013 in the Lynn Valley area. We expect a series of rehabilitation contracts similar to those described above to be developed.

We have budgeted \$150k and will be performing bulk lateral inspections in the Ross & Allen sewershed with the intent of identifying locations for lateral rehabilitation and assessing if lateral inspection can effectively identify defects that will help strategically direct the I&I reduction program.

3.7.1.5 Sanitary Sewer Inspection

Annual CCTV Inspection Contract

The 2014 CCTV contract is in progress and is focused on three principal areas;

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- substantially completing the sanitary sewers in the Lynn Valley sewershed that are difficult to access, such as those in backyard easements,
- mains in the Birchlynn sewershed that were identified as having a structural performance grade (WRc) of 3 or higher or identified as requiring rehabilitation through the 2009 CCTV inspection program, and
- Re inspecting mainline and lateral connection grouting that was performed between 2003-2006 to assess for the current condition and performance as an infiltration inhibitor.

Based on previous years' inspection results we anticipate the 2014 program will identify 60-80 locations of potential infiltration. Our rehabilitation protocol dictates that if one main pipe joint in a pipe section is identified as having active or signs of infiltration then all joints in that pipe section are pressure tested and injection grouted if they fail the pressure test. The results of this data will indicate how effective CCTV inspection is at identifying infiltrating pipe joints.

3.7.1.6 Lateral Interface Inspection and Rehabilitation

Lateral connection interface inspections occur during our annual contracted main line video inspection programs. Rehabilitation of lateral interface grouting will occur as a component of the annual sewer main rehabilitation programs. Our protocol is that if one lateral interface in a pipe section is identified as having active or possible infiltration all lateral interfaces in that pipe section are targeted for pressure testing and injection grouting. The results of this data will indicate how effective CCTV is at identifying infiltrating lateral connections.

3.7.1.7 Installation of Sanitary Lateral ICs

Staff and contractors will complete the installation of sanitary IC's on all SFR properties that have been identified as potentially being cross connected. We have identified 19 that will be completed in 2014 and expect that the dye testing program will identify more. We have approved 2014 budgeted in excess of approximately \$300k for IC installation efforts focused in the Lynn Valley area. An ongoing IC installation program is envisioned for the balance of the municipality for the future. A survey of the Lynn Valley sewershed completed in 2011 indicate that only 10% of the existing sewer lateral stock have IC's installed.

3.7.1.8 Lateral Inspection Equipment

We have approved budget of \$160k for a new main line lateral launch camera (crawler). This camera will have the ability to inspect both main line pipes and laterals launching from the main pipe. We have budgeted an additional \$25k for two a new IC launch lateral inspection (push) camera. We are researching both products for expected purchase in 2014.

3.7.1.9 Lateral Inspection Program

Lateral inspections are an integral technique being used by DNV crews to better understand results of the smoke and dye testing results and to definitively locate points of cross connection for directing correction construction works.

A new lateral CCTV inspection program will be developed and started in 2014 with the emphasis on the Lynn Valley sewershed. The approved budget of \$220k will allow for approximately 400 lateral inspections. The goal of the new lateral inspection program is to identify the effectiveness of lateral inspection as a tool for identifying points of inflow and infiltration.

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3.7.1.10 Smoke and Dye Testing Program

In 2014 the emphasis will be on dye testing all 62 SFR properties identified in the 2012/13 smoke testing program as having a potential cross connection. In 2014 we will also be performing bulk dye testing in the Birchlynn sewershed as a whole. Methods for dye testing without property line inspection chambers will be explored.

The program will continue in the Birchlynn and Ross & Allan sub sewershed with the focus on identifying cross connections. It is hoped this program will be substantially complete for the Lynn Valley catchment in 2014.

3.7.1.11 Cross-Connection Removal

Six cross connections in the Crestlynn area were removed in 2012. The next phase of cross connection remediation will be in the Birchlynn and Ross and Allen catchments, both sub-sewersheds of the Lynn Valley catchment, after the dye testing program has identified the property by property remediation requirements. We expect to complete this remediation work in 2014.

4 Conclusion

The DNV has been working on executing our I&I reduction program for a number of years. A concerted effort, with much higher levels of funding, began in 2011 to allow us to ensure we could meet the following objectives in a timely manner.

1. Eliminate the need for a MV's \$15M concrete sewage retention tank at the Lynn Branch inverted siphon,
2. Fulfill our ILWRMP responsibility by taking measures to reduce I&I and maintain our infrastructure,
3. Free up capacity in our sanitary sewer system to accommodate OCP forecasted population growth,
4. Identify local cost to benefit relationship for I&I reduction programs and effective removal rates,
5. Reduce wet weather sewage flow in anticipation of a potential revision to Metro's sewage rate structure, and
6. Deepen our understanding of DNV's long term sanitary sewer loading that will assist MV in designing the planned new North Shore Waste Water Treatment plant.

The DNV's intent is to complete I&I reduction work in the Lynn Valley sewershed that will reduce the occurrence of SSO's from the LBISB to acceptable ILWSMP levels of less than once every 5 years for 24 hour rain events.

The relationship between rainfall and inflow/infiltration is quite complex and efforts to determine an effective cost/benefit solution to reduce I&I will take time to ensure we are appropriately analyzing the many available remediation techniques against appropriate rainfall events. Cross connection removal already shows significant promise as a positive cost/benefit technique to reduce inflow. We believe this will be confirmed and additional techniques will be identified in the next few years.

Once we are able to complete cost-benefit analysis, we can determine which components will be expanded to the larger DNV area. Although early indications that addressing inflow will be more economical on a \$/L basis, compared to addressing infiltration, we expect that we will require reduction in infiltration to hit our peak flow reduction target in the Lynn Valley area. The

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cost-benefit data produced from our I&I reduction program will continue to develop and become a reliable tool for our I&I program strategy development.

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Appendix F: District of West Vancouver: Sewer Rehabilitation and Flow Monitoring Program

E-mail of March 31, 2014 (9584846)

From Andy Kwan, P.Eng., Utilities Engineer, District of West Vancouver

To Regional Engineering Advisory Committee – Liquid Waste Subcommittee

Ambleside Public-side Sewer Rehabilitation and Flow Monitoring Program

Background

Ambleside is one of the oldest utility serviced neighbourhoods in West Vancouver, bounded by 25th Street to the west, the Upper Levels Highway to the north, 11th to the east and Burrard Inlet to the south. For sanitary sewer collection purposes, the area has been delineated into six sub areas servicing an area of approximately 292 hectares. With the exception of the Citrus Wynd sewage area north of Horseshoe Bay, the District's sanitary sewer collector systems, including those within the Ambleside Area, tie into Metro Vancouver's Hollyburn Interceptor sewer which conveys sewage to the Lions Gate Treatment Plant Facility, owned and operated by Metro Vancouver. The majority of the sewers within the Ambleside area were installed in the late 1950s and early 1960s; with pipe sizes ranging between 150 mm and 250 mm in diameter and are mostly vitrified clay in material. Service connections are generally 100 mm in diameter with the pipe material being either asbestos cement or vitrified clay.

Sewer Rehabilitation Program

In 2010 the District of West Vancouver initiated a rehabilitation program with the assistance of the Building Canada Fund. There were two specific reasons for the rehabilitation program. The first reason is to repair and replace aging infrastructure to extend the service life. The second reason was to assess the effect of rehabilitation on the inflow and infiltration (I&I) rates. A consultant was engaged for contract administration, pipe condition assessment, and evaluation and implementation of a flow monitoring program for the subject area.

The project scope included CCTV, smoke testing and manhole inspection for sewers within three of the six Ambleside Sub Areas, known as C1, C2 and C4. Following pipe and system condition assessment, deficiencies were prioritized and a suitable means of rehabilitation was determined. Methods of rehabilitation included pipe bursting, relining, external point repairs and trenchless point repairs. In total, the program included pipe bursting of 1.6 km of 200 mm to 250 mm diameter sewer mains, relining of 3.2 km of sewer mains, open cut replacement of 0.1 km, the re-establishment of 164 sewer service connections, 44 point repairs and the replacement of 16 manholes. All repairs and replacement were within District of West Vancouver property with no rehabilitation completed on private properties.

A summary of the sewer rehabilitation works undertaken for the sub areas, known as C1, C2 and C4, is characterized below:

Table 10 Ambleside Sewer Rehabilitation Summary

Catchment	Area (Ha)	Sewer Mains (m)	Rehabilitation					
			Pipe Burst (m)	Reline (m)	Open cut (m)	EPR (≤ 10 m)	TPR (≤ 4 m)	Manhole (ea.)
C1	34	4 000	-	243.4	-	1	2	-
C2	110	19 370	1194.1	2675.1	81.3	21	14	14
C4	17	2 295	210.4	296.3	22	1	5	2

Flow Monitoring Program

The objective of the flow monitoring study was to record continuous flow data during dry and wet weather months to quantify the pre and post rehabilitation I&I for the three sanitary sub areas. The post rehabilitation data would be used to evaluate the extent of the I&I reduction for the rehabilitated area.

Four manhole locations were selected for flow monitoring stations based on their proximity to the sewerage areas. Although there were only three catchments, a fourth monitoring station had to be installed to quantify sewer flow contributing to the C2 catchment from catchments above the Upper Levels Highway, ensuring no large spikes from the upstream catchment adversely affected the recorded results for the C2 catchment. Data was collected continuously at 5 minute intervals and was downloaded monthly from the sites with the sites being maintained accordingly, as required.

The monitoring program commenced August 2010 and ran through April 2013. Pre- and post-rehabilitation storm events were plotted separately on the West Vancouver Municipal Hall rain gauge IDF curve. The largest pre-rehabilitation storm recorded was a 1-hour, 2 year storm event and the largest post-rehabilitation storm recorded was a 6 hour, 1 year storm event. Storms were selected for each site to help establish the pre-and post-rehabilitation I&I rates.

I&I Analysis

Dry and wet weather flows were combined with rainfall data to develop and assess I&I in terms of Rainfall Dependent Inflow and Infiltration (RDI&I), an analysis methodology developed by the Greater Vancouver Sewerage and Drainage District's I&I task force.

The RDI&I method expresses sewer flow events in litres/hectare/day (L/ha·d) versus rainfall volume and allows for a comparison of the RDI&I volume versus the desired volume of 11,200 L/ha·d. Maximum 24-hour RDI&I volume is determined and the corresponding maximum 24 hour rainfall volume is calculated for each storm event. These data points are plotted on an RDI&I envelope evaluation graph and a line is constructed to fit the points. The line is then extrapolated to intersect with the 5-year rainfall volume, allowing for an average 5-year storm event RDI&I volume to be determined.

The following table represents the maximum RDI&I rates in L/ha·d calculated for each of the sites for pre- and post-rehabilitation storm events measured during the monitoring program.

Table 11 Ambleside Estimated Peak RDI&I

Site	Catchment	Area (ha)	RDI&I (L/ha·d)		
			Pre-Rehab	Post-Rehab	Reduction (%)
2	C4	17	22 500	17 500	22%
3	C1	34	16 000	14 500	3%
4	C2	110	34 000	23 000	32%

**maximum RDI&I rate assessed for a projected 24-hour, 5 year return storm event*

Program Cost Evaluation Summary

Table 12 below summarizes the costs associated with the Ambleside Rehabilitation and Flow monitoring program.

Table 12 Ambleside Rehabilitation Program Cost Summary

Description	Cost
CCTV	\$ 152,987.42
Flow Monitoring	\$ 57,490.00
Pipe Bursting	\$ 550,524.35
Relining	\$ 531,703.34
Service Conn. Re-establishment	\$ 254,271.80
TPR	\$ 31,931.58
EPR	\$ 140,500.00
Open Cut	\$ 24,200.00
Manhole	\$ 98,375.79
Consultant	\$ 149,510.00
TOTAL	\$ 1,991,494.29

Table 13 below compares catchments, total length of sewer main and percentage of rehab in linear meters, with the cost expenditure per catchment versus I&I reduction. Assuming program costs are weighted equally per meter of rehab within the catchment, the average cost per meter of rehab is \$421.70. We can conclude that pipe rehabilitation methods such as pipe bursting and relining have a positive effect on I & I reduction. However, we did not find any correlation between percentage I&I reduction versus the percentage rehab within the catchment. Further, although we have repaired or replaced all visible defects found in the mains on the public side of the sewer system, we were unable to achieve a reduction to the Metro Vancouver desired volume of 11,200 L/ha·d.

Table 13 Ambleside Rehabilitation Program Cost per meter Versus I&I Reduction

Catchment	Area (ha)	Sewer Mains (m)	Sewer Main Rehab - (combined, m)	% Total Rehab	Cost as % Total Rehab	I&I Reduction (%)
C1	34	4 000	243.4	5%	\$ 102,642.59	3%
C2	110	19 370	3 950.4	84%	\$ 1,665,896.85	32%
C4	17	2 295	528.7	11%	\$ 222,954.55	22%
TOTAL	161	25 665	4 722.5	100%	\$ 1,991,494.00	n/a

* assuming costs are weighted equally for the percentage rehab/catchment in meters yields a total average program cost of \$421.70/m rehab.